

Widespread ecological responses and cascading effects of the 2021 western North American heatwave

Received: 30 November 2024

Accepted: 15 January 2026

Published online: 11 March 2026

 Check for updates

Julia K. Baum¹✉, Margaret A. Slein^{2,3}, Josef C. Garen^{2,4}, Zihaoan Sang⁵, Sandra Emry^{2,3}, Katie J. A. Goodwin^{2,4}, Courtney G. Collins^{2,4,6}, Jayme Lewthwaite⁷, Michelle Tseng^{2,3,4}, Sean T. Michaletz^{2,4}, A. Cole Burton^{2,8}, Christopher D. G. Harley^{2,3,9}, Diane S. Srivastava^{2,3} & The 2021 Western North American Heat Dome Data Consortium*

Extreme weather events are increasing in frequency and intensity, but their ecological impacts remain less well understood than those of gradual climate change, largely owing to the challenge of studying unpredictable, short-lived events. The 2021 western North American heatwave is among the most extreme on record globally, yet a broad assessment of its ecological consequences is lacking. Here we synthesize meteorological, ecological, hydrological and wildfire data, along with process-based modelling, to quantify the heatwave and its impacts across the region. Our meta-analysis of 32 terrestrial and marine taxa reveals that over 75% were negatively impacted, but species responses ranged widely, from 99% declines to 89% increases. This variability reflects differences in organisms' thermal sensitivities, response capacities and exposures, with the latter dependent on geography, microclimate and refugia. Impacts tended to be greater for sessile marine invertebrates, algae and plants than for birds and mammals. At the ecosystem scale, changes in gross primary productivity ranged from 30% increases in cooler, wetter areas to 75% decreases in warmer, arid ones. Streamflow from snow and ice melt increased 40% during the heatwave before dropping below average, whereas wildfire activity surged 37% during the heatwave and 395% the following week. Our results underscore the urgent need for enhanced coordinated approaches to predict, detect and manage increasing heatwaves.

Extreme weather events that are amplified by climate change, including heatwaves, floods and droughts, are now major drivers of ecological change^{1,2}. These events are impacting biodiversity and ecosystem function directly through mass mortalities, local extinctions, reproductive failure, species range shifts and invasions, as well as through cascading effects mediated by species interactions and ecosystem feedbacks^{3–7}. Ecological impacts can be compounded when extreme events occur concurrently owing to multiple interacting climate drivers, such as when

heat and drought conditions lead to wildfires^{5,8,9}. Further, while extreme weather events are often short in duration, their legacy can persist for years because of genetic diversity losses, the loss or introduction of new species, prolonged population recovery times or failures to recover, such as when ecosystems shift to alternate stable states^{3,5,10}. Increased monitoring, understanding and prediction of the species and localities most vulnerable to extreme weather impacts is imperative if we are to effectively manage ecological systems this century. However, despite

A full list of affiliations appears at the end of the paper. ✉e-mail: baum@uvic.ca

the increase in extreme terrestrial and marine climatic events over the past decade, assessing the ecological impacts of short-duration extreme events remains challenging because of limited forewarning of their occurrence and the narrow window of time within which to conduct studies.

Ecological responses to heatwaves—which are ultimately driven by changes in the performance and survival of individual organisms—can vary widely because differences in thermal stress exposure, organismal sensitivity and response capacity^{11,12} create a mosaic of thermal vulnerability among individuals and species across the landscape^{13–15}. Exposure can vary enormously across space owing to large-scale atmospheric condition differences; intermediate-scale factors including variation in topography, humidity and wind; and fine-scale microclimatic differences^{13–17}. Exposure among individuals can also vary depending on organismal traits (for example, size, shape and surface optical properties) and metabolic rate, which help to mediate the difference between ambient temperature and body temperature^{18,19}. Thermal stress also includes vapour pressure deficits (VPDs; the evaporative demand of the atmosphere), which increase nonlinearly with temperature and can thus vary widely during heatwaves. High VPD may enhance heat losses from organisms that rely on evaporative cooling for thermoregulation, provided that water is available, or lead to desiccation, injury and mortality when water is unavailable^{6,7}. Heatwave sensitivity depends on physiological tolerance limits, which vary among species, individuals and even life stages, while response capacity may depend on physiological or behavioural thermoregulation. Evaporation is a common mechanism of physiological thermoregulation, including in animals that sweat and plants that transpire^{20–22}. Behavioural thermoregulation, widely used by ectotherms and endotherms alike^{23,24}, includes energy conservation during hot periods and shifts in timing or locations to select cooler periods or microclimates within the organism's heterogeneous 'landscape of heat'¹³. For cold-adapted northern species, behavioural thermoregulation has been better studied for cold weather (for example, refs. 25,26) than for the high temperatures expected under climate change^{27–29}, especially heatwaves.

The 2021 western North American heatwave (25 June to 2 July 2021)—commonly referred to as the 'heat dome'—not only shattered regional temperature records but is one of the most intense climate change-fuelled heatwaves on record globally³⁰. Extreme temperature anomalies blanketed much of the western part of the continent during the event, with temperatures soaring over 10 °C above long-term average levels. The Canadian temperature record was broken, with the town of Lytton, British Columbia (BC), reaching 49.6 °C before tragically burning in a wildfire. This is reportedly the hottest temperature ever recorded north of 45° latitude or anywhere in Europe or South America^{31,32}. Indeed, with temperature extremes reaching a remarkable four standard deviations (s.d.) above regional long-term averages, this event ranks as the sixth most extreme heatwave since 1960 in the world³⁰. Ecological insights from previous, larger heatwaves are limited in part because they took place decades ago, were of limited duration or came with little forewarning. Attribution studies suggest that climate change made the 2021 western North American heatwave 150 times more likely and that with 2 °C of planetary warming, heatwaves as strong or stronger could become a decadal occurrence^{33,34}. Given this, understanding the ecological impacts of extreme heatwaves is critically important. Yet, whereas the impacts of major heatwaves on human health and society are often well documented^{35–37}, much less is known about how other species, and their ecosystems, are affected by such extreme events.

Here, we use a mixed-methods approach, leveraging field data from scientists working in the region affected by the 2021 western North American heatwave, remote-sensing and open access datasets, media reports and process-based modelling to advance understanding of how organisms and ecosystems respond to short-term extreme temperature anomalies. We first demonstrate spatial variability in exposure for three different meteorological variables (air temperature, land surface temperature and VPD) across the heatwave's geographic extent

and combine these records with species' thermal performance curves to show how the same extreme climatic event could affect the performance of different organisms in distinct ways. We then conducted a meta-analysis of published and unpublished datasets from long-term monitoring programs, well-timed experiments or opportunistic observations quantifying ecological responses to the heatwave across taxa. We hypothesized that across the heatwave's overall negative effects on organisms, sessile organisms would be more negatively impacted than motile ones owing to their inability to access cooler microclimates and that species responses would range from immediate but potentially short-lived behavioural responses (for example, change in activity levels during the heatwave) to mortality and other effects (for example, reproductive failure) with longer-lasting demographic effects. Complementing this meta-analysis with additional literature (that did not meet the criteria for the meta-analysis), field observations and media records, we further document the breadth of ecological responses to the heatwave. Finally, we used process-based models, open-access government datasets and satellite-derived data to quantify the heatwave's broader ecosystem impacts: changes in gross primary productivity, streamflow and wildfire activity. Our synthesis provides insights into the breadth and variability of extreme heatwave impacts, and factors driving this variability, and ends with recommendations for scientists and policymakers to better prepare for future heatwave events.

Extreme, but variable, climatic conditions

The 2021 western North American heatwave, which was the most intense between 25 June and 2 July (Extended Data Fig. 1), covered an area of ~3 million km², spanning almost 27° of latitude, from northern California (38.82° N), USA, to the Northwest Territories, Canada (65.46° N) (Fig. 1). From the west coast, the heatwave reached eastwards across BC, Alberta and Saskatchewan into west-central Manitoba in Canada and across Washington and Idaho into western Montana in the USA (Fig. 1). Across this mid-latitude region, in which the average historical daily maximum air temperature (during the same days as the heatwave) was 21.2 °C, temperatures rose to a mean daily maximum of 30.5 °C (Fig. 1a). Temperature records are normally broken by a degree or two, but during this heatwave, the median extreme daily temperature anomaly was 14 °C. Most locations therefore experienced temperatures over 35 °C at least once, and numerous locations were subject to even greater anomalies (20–30 °C; Fig. 1a and Extended Data Fig. 2a). Land surface temperatures soared to even higher daily maximum levels than air temperatures across the affected region, reaching over 50 °C in some locations (Fig. 1b), while VPD doubled on average relative to their historic levels (Fig. 1c).

Despite the heatwave's extremity, the operative temperatures that organisms experienced varied widely depending on their specific location and microclimate (Fig. 1 and Extended Data Fig. 2). Operative temperatures are the outcome of several organismal and environmental heat fluxes and how these are mediated by organismal energy balance traits^{19,20}. While operative temperatures almost never equal the microclimate conditions measured for the land surface or air at 2 m elevation, these are still useful proxies for characterizing microclimate variation. Indeed, these climatic measures, as well as VPD, all varied enormously across the heatwave's geographic range (Fig. 1 and Extended Data Fig. 2). Air temperature anomalies were extremely high throughout much of BC and western Alberta, while extremely high land surface temperature anomalies were observed in the Canadian Rocky Mountains, Cascade Mountains and low elevation arid grassland and desert areas of the interior of BC (Extended Data Fig. 2a,b). VPDs were unusually high in Oregon and Washington states (Fig. 1c and Extended Data Fig. 2c). Within these larger regions, we also expect biologically relevant fine-scale variation in thermal exposure depending on local topography, forest cover and other factors. For example, temperature loggers on remote cameras in the BC central interior

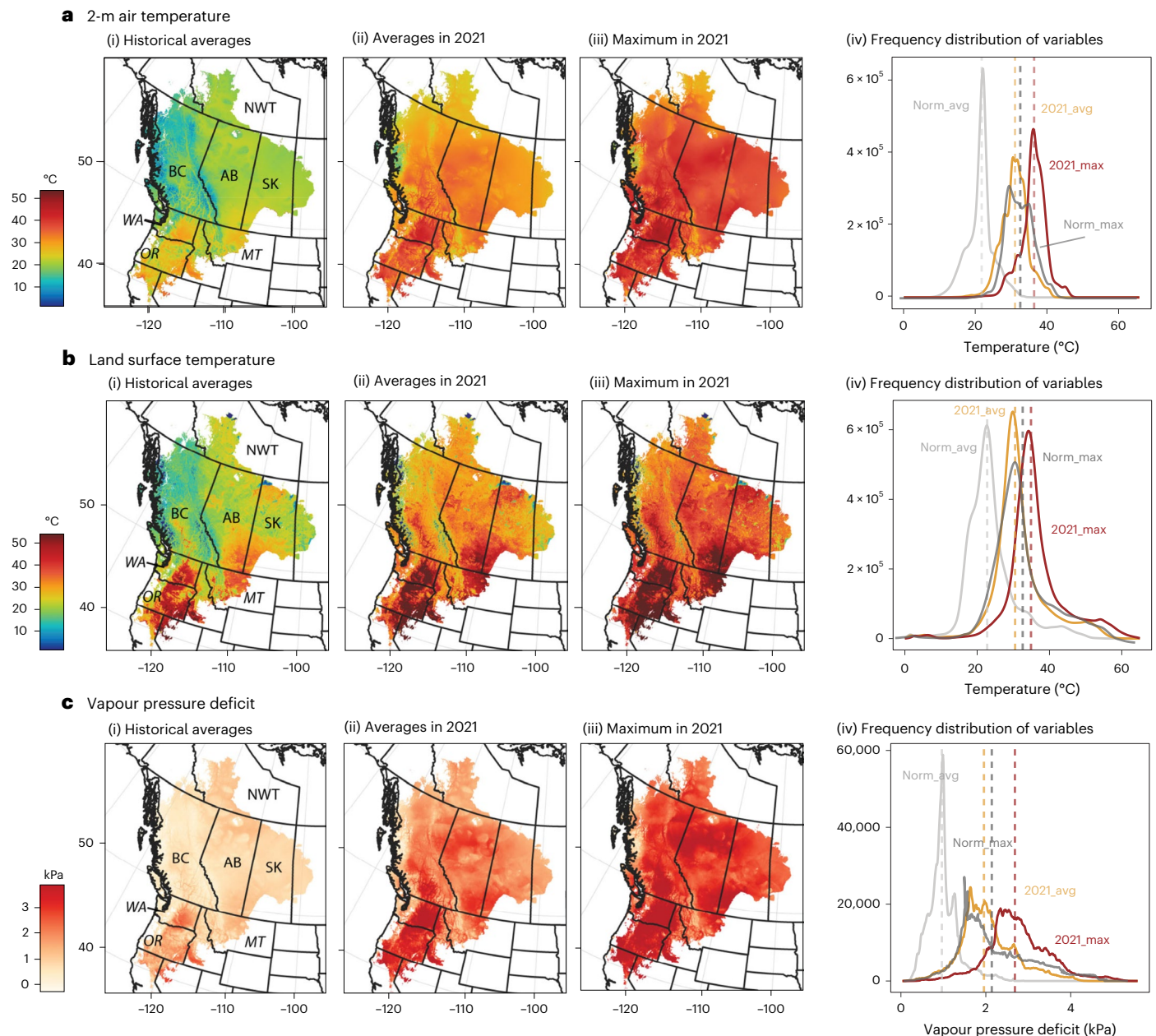


Fig. 1 | Extreme climatic conditions during Western North America's 2021 heatwave. a–c, The 2-m daily maximum air temperature (a), daily daytime land surface temperature (b) and heat-induced VPD (c) across the impacted region. For each climatic variable, the (i) historical daily average for the same days as the 2021 heatwave, 25 June to 2 July, (ii) daily average values and (iii) daily extreme

values of temperatures for the 2021 heatwave period are shown. Frequency distributions of each variable are shown in (iv), with dashed vertical lines showing the median values. On the basis of data availability, the historical period for **a** and **c** was 1980–2020 and for **b** was 2000–2020; see the Methods for details.

showed that noon temperatures recorded at individual sampling sites ranged from a low of 16.7 °C to a high of 46 °C, depending on local sun exposure, forest cover and elevation (Extended Data Fig. 3). On coastal rocky shorelines, the land surface and even organismal shell surface temperatures reached extremes in excess of 50 °C during the heatwave³², which is well above typical summer temperatures for the region³⁸.

Consequences of heat stress exposure depend on organismal sensitivity

How organisms responded to the 2021 heatwave depended not only on their exposure but also on their physiological temperature sensitivity³⁹. As basic thermal performance parameters such as extreme tolerance limits remain unmeasured for the vast majority of species,

we developed a conceptual model to illustrate how operative temperatures and organismal thermal performance curves³⁹ interact to create a diversity of organismal responses to heatwaves (Fig. 2). We simulated thermal performance curves for two hypothetical species that differ in their optimal temperatures and breadth of thermal performance (Fig. 2a). When the operative temperatures of each species are taken as distributions representing historical average land surface temperatures versus land surface temperatures from the 2021 heatwave (Fig. 2b, blue and red distributions), differing performance outcomes result. Under temperatures experienced during the heatwave, the species with the higher optimal temperature and wider performance breadth experienced an overall increase in performance and contraction of its performance distribution as temperatures approached its optimal temperature (Fig. 2d). By contrast, the species with the lower

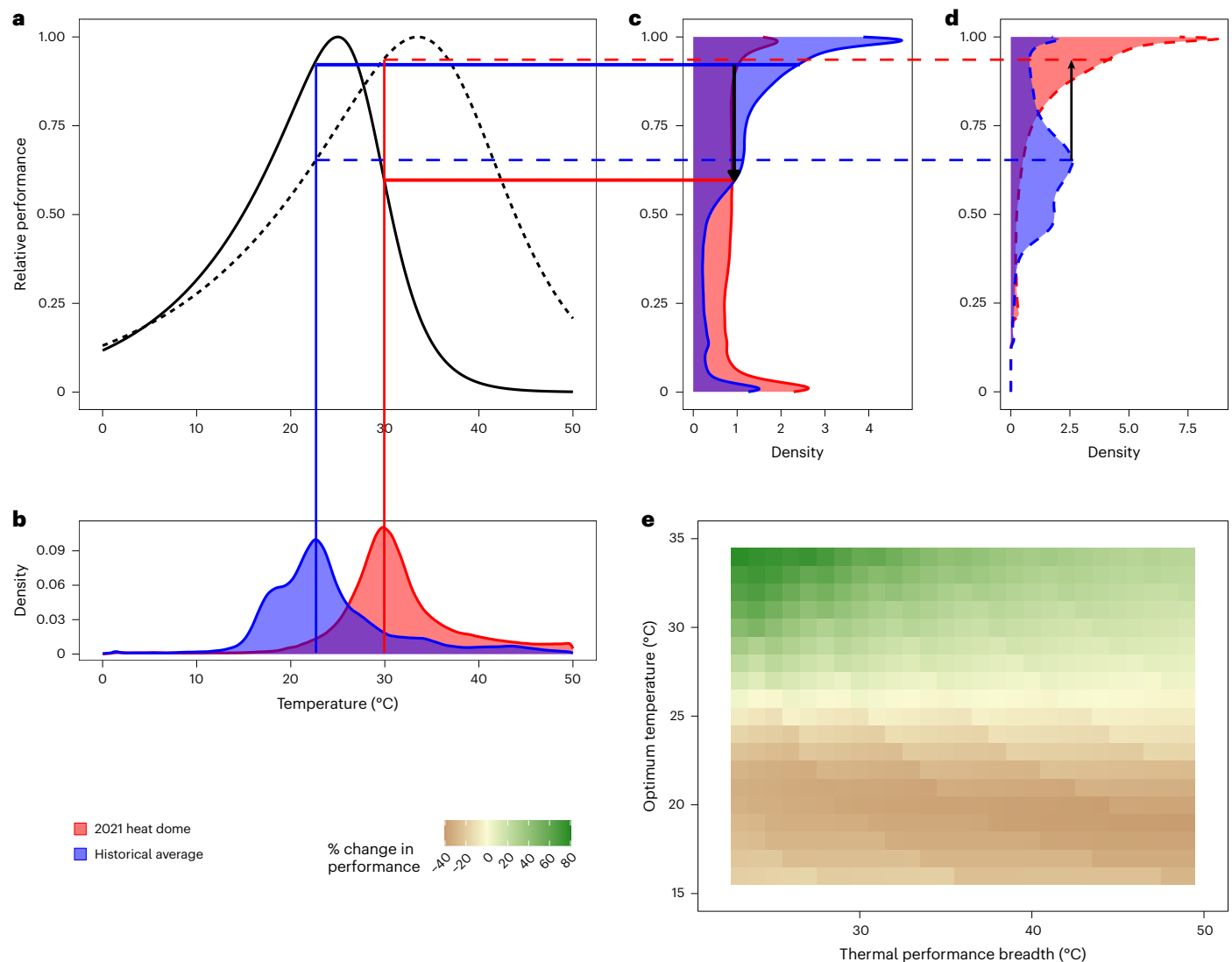


Fig. 2 | Organismal operating temperature and thermal performance jointly determine total performance. **a**, Example thermal performance curves representing two contrasting hypothetical species, one with low temperature optimum and narrow thermal breadth (solid black line) and one with high temperature optimum and wide thermal breadth (dashed black line). **b**, Estimated land surface temperature distributions during the period of the 2021 western North American heatwave (25 June to 2 July; red) and during the historical average (1980–2020) for the corresponding days (blue). **c–d**, Performance distributions of the hypothetical species from **a** when exposed

to the temperature distributions from **b**. The figure shows the performance change (blue area compared with the red area) experienced by the first species (solid line) (**c**) and that experienced by the second species (dashed line) (**d**). Blue and red lines connecting **a** to **d** show the relationship between the temperature distribution and performance distributions at the most commonly observed historical and 2021 heatwave temperatures, as mediated by the thermal performance curves. **e**, Heat map showing the relationship between properties of the thermal performance curve (breadth and optimum temperature) and relative organismal performance.

optimal temperature and narrower performance breadth experienced an overall decline in performance (Fig. 2c), as temperatures during the heatwave generally exceeded its optimal temperature. The distribution of performance values for this species was much wider and more evenly distributed during the heatwave (Fig. 2c) since its performance declined rapidly as temperatures increase above the optimal value (Fig. 2a). For both species, variability in performance (Fig. 2c–d) reflected the spatial variability in temperature exposure and location of the temperature distribution relative to each species' optimal temperature. Overall, performance during a heatwave depends most strongly on species' optimal temperatures, with species that have low optimal temperatures experiencing strongly negative performance changes regardless of breadth and species, with high optimal temperatures showing generally positive responses (Fig. 2e). Thermal breadth primarily influences the magnitudes of these effects, increasing performance in species with high optimal temperatures but having relatively little influence

on species with low optimal temperatures. As this model illustrates, widely variable ecological effects are predicted owing to the interactive effects of differences in organismal physiology, spatial variation in temperatures and extremity of temperature increase. Further differences are anticipated on the basis of other factors, including behavioural responses to temperature change.

Widespread but variable organismal responses to the heatwave

Even with highly variable temperature anomalies and organismal sensitivities, the meta-analysis of species' responses to the heatwave revealed a strong negative overall effect of the event (intercept-only model; log response ratio -0.91 , 95% confidence interval (CI) -1.42 to -0.39 ; Fig. 3a and Extended Data Table 1). Over three-quarters ($n = 25/32$) of taxa responded negatively to the heatwave (Fig. 3b), with most of the largest effects detected being striking decreases in

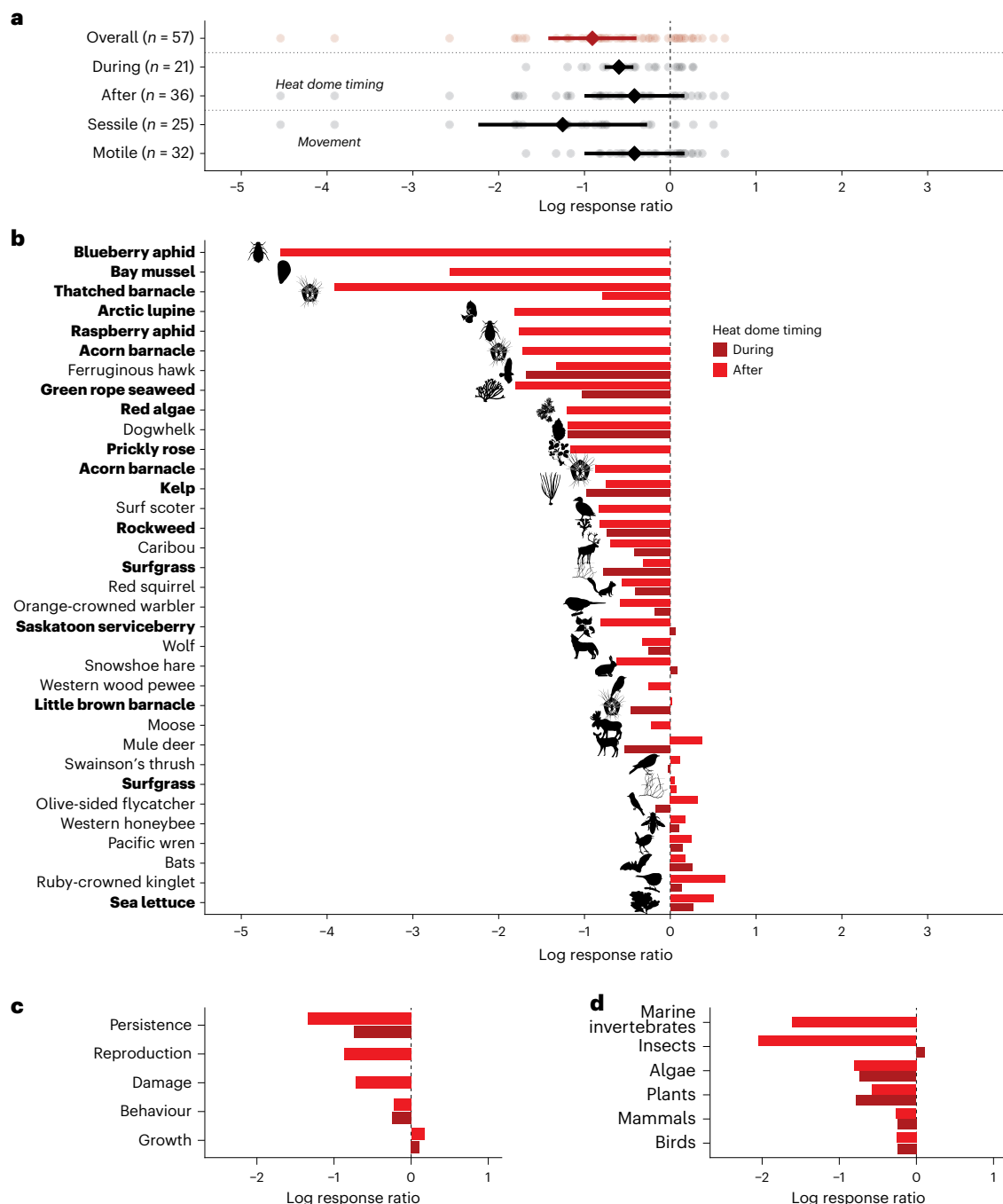


Fig. 3 | Ecological responses to the 2021 heatwave. a, Estimates from a meta-analytic model with moderators from the final model (time frame and movement). Model estimates (black diamonds) are presented with 95% CIs (horizontal error bars) and raw effect sizes (grey points), with an additional overall estimate from an intercept only model (red diamonds) and raw effects sizes (red points). **b–d**, Individual response ratios ($n = 57$ responses) for each species (group) ($n = 32$ different taxa; note that acorn barnacle and surfgrass are

included twice, as data for them were obtained from two independent studies) (**b**), each response type (persistence included changes in abundance or survival) (**c**) and each taxonomic group (**d**). Dark and light red bars denote data collected during versus after the heatwave, respectively. Bolded species in **b** indicate sessile organisms, whereas nonbolded species indicate motile organisms. Details of individual datasets are presented in Extended Data Table 2.

population numbers (measured as abundance or survival) or reproduction (Fig. 3c). Intertidal invertebrates (*Balanus glandula*, *Chthamalus dalli*, *Mytilus trossulus* and others) baked by the billions, green rope seaweed (*Acrosiphonia* spp.) bleached and vanished, aphids (*Ericaphis fimbriata* and *Amphorophora agathonica*) died stranded on their blueberry and raspberry plant hosts and the flowers of Arctic lupines (*Lupinus latifolius*) wilted on their stems (Fig. 3b,d and Extended Data Table 2). By contrast, behavioural responses of birds and mammals (that is,

changes in songs or calls, or in camera trap detections as proxies for mammal movements), and growth rate responses of some insects, were often less negative, neutral or even mildly positive (Figs. 3b–d and 4, Extended Data Fig. 4 and Extended Data Table 2). Ferruginous hawks (*Buteo regalis*), whose flight levels declined by 81% during the heatwave, were a striking exception (Fig. 3b), possibly because they are an open prairie bird and would have had high exposure to the heatwave. Amid this variability, only the timing of study responses was statistically



Fig. 4 | Organismal responses to the 2021 heatwave. **a–c**, Camera trap detections of mammals in or near Itcha Ilgachuz Provincial Park, including mountain goat (*Oreamnos americanus*) experiencing unusual heat at high elevation, 27 June 2021 (**a**), moose (*Alces alces*) cow and calf in a cooler site shaded by forest canopy cover, 30 June 2021 (**b**) and grey wolf (*Canis lupus*) in an exposed site with little canopy, 26 June 2021 (**c**). **d–f**, Mortality of marine invertebrates, including dead mussels (*Mytilus trossulus*) in Lighthouse Park, West Vancouver, 28 June 2021. Note that there is still meat in the shells, indicating that they died very recently (**d**), dead sea star (*Pisaster ochraceus*) in Lighthouse Park, West Vancouver, 28 June 2021 (**e**) and recently dead green false-jingles (*Pododesmus macrochisma*) photographed at Elliotts Beach Park near Ladysmith, BC, 10 July

2021. Shells are ~6–8 cm in diameter. Note the attachment scars visible in the centre and lower left of the image (**f**). **g–i**, Leaf scorch in plants, including salal (*Gaultheria shallon*) leaves in Malcolm Knapp Research Forest, Maple Ridge, BC, 8 July 2021 (**g**), extensive leaf scorch still present in kinnikinnick (*Arctostaphylos uva-ursi*) 1 year after the heatwave, 1 June 2022, in Southern Interior, BC (**h**) and leaf scorch in coastal Douglas-fir (*Pseudotsuga menziesii*) and western hemlock (*Tsuga heterophylla*) plantation forests in western Oregon and Washington (**i**). Species in **b**, **c** and **d** were included in the meta-analysis, while responses of those that could not be included owing to insufficient data are presented in Extended Data Table 3. Photo credits: **a–c**, UBC WildCo Lab, Government of British Columbia; **d–f**, C.D.G.H.; **g**, D.S.S.; **h**, S.D.-H.; **i**, Gabriela Ritokova.

significant, with more negative effects detected during, compared with after, the heatwave (log response ratio -0.18 , 95% CI -0.35 to -0.01) (Fig. 3a and Extended Data Table 1), probably reflecting the transient behavioural responses of motile species.

Responses to the heatwave differed substantially with organismal motility, with sessile organisms unable to shelter and thus experiencing greater losses, although not statistically significantly different from motile organisms (log response ratio -0.84 , 95% CI -1.82 to 0.15 ; Fig. 3a,b and Extended Data Table 1). Sessile organisms in exposed areas suffered extensive mortality (mussels (*Mytilus trossulus*) 92%, barnacles (*Semibalanus cariosus*) 90%) or reproductive failure (lupines 84%) (Figs. 3b and 4), and in all cases where the performance of sessile organisms could be directly compared spatially between heatwave-exposed locations and nearby sheltered ones (for example, shaded versus unshaded), species exhibited substantially reduced performance in the more exposed area (Fig. 3b and Extended Data Table 2). The importance of these stark differences in thermal stress exposure for sessile organisms is further underscored by the contrasting impacts of the heatwave on surfgrass (*Phyllospadix scouleri*) in different parts of BC: along southern Vancouver Island, where low tides occurred at midday during the heatwave, surfgrass in the intertidal zone was exposed to the hottest local temperatures and declined drastically following the

heatwave ($\sim 26\%$; Fig. 3b and Extended Data Fig. 5), whereas further north on the Central Coast of BC, where low tides occurred roughly 2 h earlier and were not as low at midday, surfgrass was buffered from the worst heat and experienced only minimal losses (5%; Fig. 3b).

In addition to the responses included in the meta-analysis, we summarize observations from peer-reviewed literature and qualitative observations from vetted field reports and well-documented media accounts. These observations on rockweed, seagrass, trees, shrubs and other organisms corroborated these findings on motility and further highlighted that otherwise motile species could also be trapped by the heat (Fig. 4 and Extended Data Table 3). The latter included organisms at vulnerable life stages, with nestlings plummeting to their deaths from overheating nests⁴⁰, or in inescapable habitats, such as salmonids in small streams that suffocated (Extended Data Table 3).

We also documented several effects of the 2021 heatwave that reflect altered species interactions. These examples include quantitative data as well as qualitative observations that describe impacts not represented in the empirical datasets. In the rocky intertidal, seaweed and surfgrass losses appear to have facilitated the observed gains in sea lettuce (*Ulva* spp.), an early-successional opportunistic green algae (Fig. 3b). In the interior of BC, wolves (*Canis lupus*) slightly increased their movements during the heatwave, potentially as a response to

maintain encounters with select prey species (that is, moose *Alces alces*, mule deer *Odocoileus hemionus* and caribou *Rangifer tarandus*) that had significantly decreased their movements in the heat (Fig. 3b). In orchards, fruit fly larvae cooked inside scorched cherries, whereas on the coast, high sea surface temperatures led to bacterial contamination of shellfish, which cascaded to infect dozens of people with vibriosis (Extended Data Table 3).

Effects on ecosystem function

Modelling changes to a key terrestrial ecosystem flux, gross primary production (GPP), we found that the increased air temperatures, land surface temperatures and VPDs experienced during the heatwave resulted in a loss of GPP for over half of the affected land area (58%), with a mean loss of 10.9% compared with historical averages (Fig. 5). However, impacts of the heatwave on GPP varied widely across the landscape, ranging from declines of 75% (5th percentile) up to increases of 30% (95th percentile) relative to historical averages (Fig. 5). GPP generally decreased in warmer and more arid regions (including the Columbia Plateau of eastern Oregon and Washington, the Interior Plateau of BC and the prairies of southeastern Alberta and Montana), where leaves were warmed to temperatures above photosynthetic optima and where VPDs were high (Figs. 1 and 5). By contrast, GPP tended to increase in cooler and wetter coastal and montane regions, particularly in BC, where leaves were warmed to temperatures closer to their photosynthetic optima and where VPDs remained relatively low (Figs. 1 and 5).

Cascading ecosystem effects of the heatwave

The 2021 heatwave also had cascading effects on large-scale ecosystem dynamics, including triggering increasing rates of summer wildfires. The heatwave was formed by a large-amplitude, high-pressure tropospheric ridge that blocked atmospheric circulation⁴¹, a meteorological phenomenon that rapidly dried dead fine fuels and has been linked to increased forest fire activity^{42,43}. Combined with the drought-like conditions⁴⁴ of early summer 2021, this created record-breaking fire weather conditions^{45,46}. Although wildfire activity was low and within historical (2000–2020) levels at the beginning of the season, the onset of the heatwave initiated an unusually active fire season (Fig. 6a), with the area of active fires far exceeding the historical averages and variation for early July. Wildfire activity climbed rapidly during the first week of the 2021 heatwave (25 June to 2 July), reaching levels 37% higher than the 20-year historical average for the same dates (but not statistically different from normal variation; Fig. 6a; $t = -0.921$, $P = 0.37$, d.f. 20). By the following week (3 July to 9 July), wildfire activity was 395% higher than the 20-year average, a statistically significant difference (Fig. 6a; $t = -5.05$, $P < 10^{-5}$, d.f. 20).

Intense heat from this heatwave also altered seasonal streamflow patterns (Fig. 6b(i)). Summer streamflow in BC is normally fed by the melting of montane snow and ice early in the summer in mid- to high-elevation watersheds and sustained by glacier melt into late summer⁴⁷. During the 2021 heatwave, streamflow in such watersheds across BC was on average 40% higher than in the previous two decades ($t_{205} = 10.1$, $P < 10^{-16}$; Fig. 6b(ii)). Streamflow was particularly enhanced in watersheds with high snow load at the start of the heatwave and (to a lesser extent) more glaciers and for streams located where the air temperatures increased the most during the heatwave relative to the previous week (Extended Data Table 4). By contrast, during August 2021, streamflow was on average 18% lower than the previous two decades ($t_{204} = 12.2$, $P = 10^{-16}$; Fig. 6b(iii)). Low streamflow was pronounced where the maximum surface temperature increased the most relative to the past two decades and in watersheds with little snow, ice or glacial cover at the start of the heatwave (Extended Data Table 4). This suggests that the heatwave, by melting the snowpack earlier and faster than normal, led to an early-summer surge in water levels in snowmelt streams and a late-summer deficit, corroborating the qualitative analysis of White

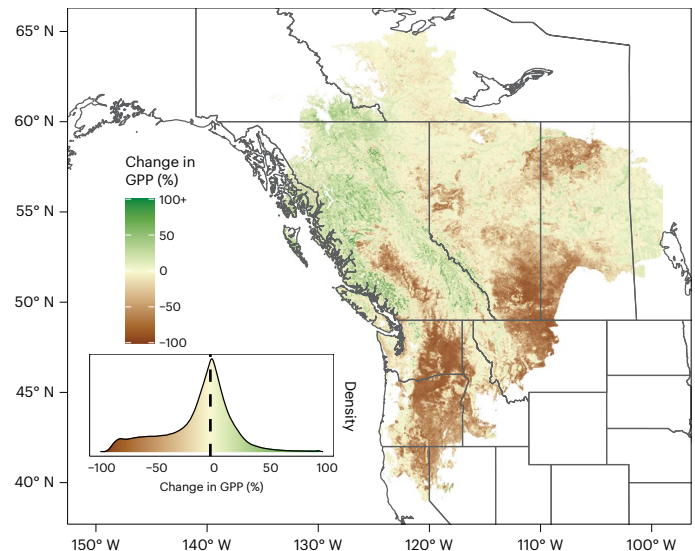


Fig. 5 | Estimated changes in a critical ecosystem function, GPP, during the 2021 North American heatwave. Modelled percentage change in GPP (that is, the total rate of gross carbon dioxide uptake by plants per unit land area) during the 2021 heatwave relative to the historical average. Green represents increases in GPP, and brown represents decreases in GPP. Inset plot shows frequency distribution of percentage change in GPP.

et al.³². In 2021, glacier melt dampened the late summer deficit, but glacier cover is shrinking across western North America, leading to long-term reductions in August stream flow^{47,48}. By contrast, streams without a snowmelt or glacial influence are subject to more intense heating without the buffering effect of elevated flows.

Discussion

Overcoming many challenges associated with documenting the ecological impacts of short-lived, extreme climatic events⁴⁹, our synthesis reveals that the 2021 western North American heatwave negatively impacted a broad array of terrestrial and marine taxa. Underlying the overall negative ecological signature of the event, the highly variable responses that we detected among taxa underscore that even during extreme climatic events, variation in species' exposures, sensitivity and response capacities can still lead to vastly different outcomes^{11,12}. The magnitude of thermal and evaporative stress anomalies to which species were exposed during this extreme heatwave varied widely ($>20^{\circ}\text{C}$ and 1–3 kPa) depending on location (Fig. 1 and Extended Data Fig. 2) and were further modified by the mosaic of microclimates created by local topography and substrates. Highly variable exposures were then met with species' variable intrinsic sensitivity to heatwaves, as illustrated by our models of organismal thermal performance curves over the range of temperatures during this heatwave (Fig. 2). At such high temperatures, our models also revealed that species had a much greater likelihood of decreased performance in comparison to previous models of risks posed by warming events⁵⁰.

Our synthesis of ecological responses to the heatwave revealed that most severely affected organisms were ones with little to no capacity to move to cooler microhabitats. These included not only sessile species but ones that are functionally sessile (for example, apterous aphids limited to single host plants and vulnerable life stages of otherwise motile species) or for which no refuge was available. Here, we distinguish quantitative effect size estimates from qualitative observational reports, which we included to increase taxonomic scope rather than infer magnitudes of effects. Some of the most negative effects of heatwaves in recent decades have been found for sessile organisms^{49,51}, including foundation species on land and in the sea^{52,53}. Our contrasting findings for birds, from muted changes in songbird

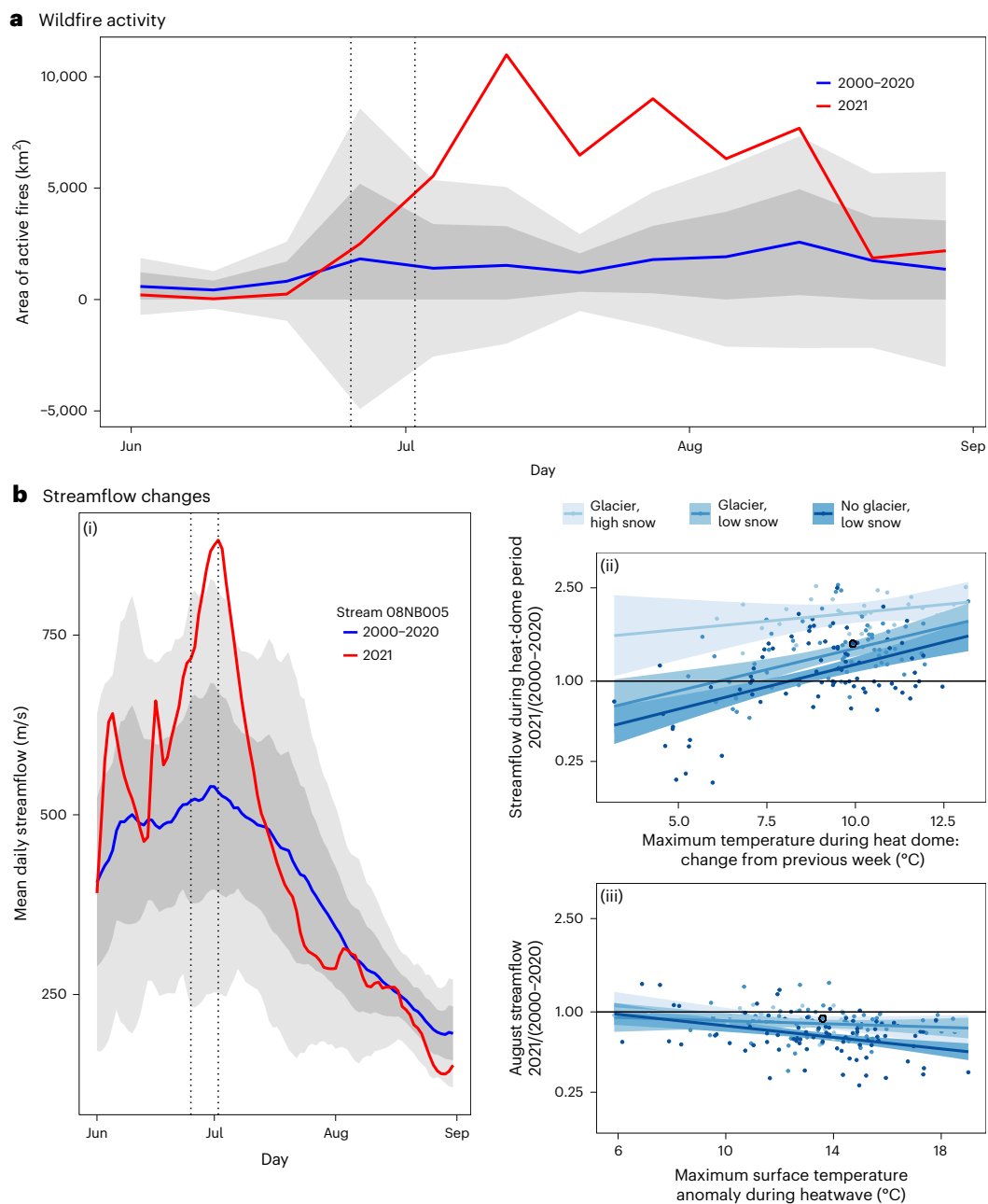


Fig. 6 | Compounding events arising from the 2021 heatwave. a, Total area of active fires for the areas of western North America that experienced the heatwave in 2021 (red line) compared with the average for the same area for the previous 20 years (blue line), with its 1 s.d. (dark-grey shading) and 2 s.d. (light-grey shading). Active fire area was identified at a 1-km spatial resolution and 8-day temporal resolution. **b**, Streamflow in areas of BC that experienced the 2021 heatwave relative to a baseline of the preceding two decades ($n = 212$ streams total). (i) Mean daily streamflow in an exemplar stream in 2021 (red line), and averaged for the baseline period (blue line), with its 1 s.d. (dark-grey shading) and

2 s.d. (light-grey shading). The ratio of these means for each stream is plotted on a logarithmic scale in (ii), against the stream's increase in maximum temperature during the heatwave relative to the previous week (blue shaded areas are 95% CIs). A similar ratio is calculated for the month of August and plotted as the y-variable in (iii). In this case, the temperature anomaly (difference from previous years) was the best explanatory variable. The exemplar stream in (i) is circled in both (ii) and (iii). Dashed lines in **a** and **b** indicate the 'heat dome period' (25 June to 2 July).

activity to nestling injuries and mortalities, underscores that heatwave impacts can vary substantially across life history stages. The meta-analysis also demonstrated that heatwave impacts can be shaped by species' behaviour. Honey bees, for example, were largely unaffected (Extended Data Fig. 4), probably owing to their highly evolved hive-cooling behaviours^{54,55} (and possibly beekeeper interventions), and less frequent detections of mammals including moose and caribou are consistent with predicted behavioural thermoregulation, such as reducing activity to conserve energy or temporarily moving to cooler

areas (for example, riparian forests and higher elevations)^{28,56}. On the whole, behavioural responses were weaker and more variable than demographic ones, probably reflecting the complexity of behavioural thermoregulation in motile species. Mammals and birds, for example, may show a wide variety of responses to heat stress, including both an increase and decrease in activity^{13,29}. The resulting behaviours reflect complex trade-offs among many interacting influences that may constrain responses to climate events (for example, predation risk, foraging, competition and social interactions)⁵⁷. Finally, there may have been

little previous opportunity for selection of plastic responses to extreme climate events, given the rarity of these events or that this level of plasticity cannot be sustained over many days given the energetic costs⁵⁸.

This study focused on immediate heatwave effects, but the documented responses suggest multiple means by which extreme climate events can generate longer-lasting effects. First, strong negative effects on survival and reproduction suggest long-term population consequences. Although some fast-growing species such as acorn barnacles recovered quickly, species with slightly slower population growth rates (rockweed and blue mussels) required a couple of years to approach pre-heatwave densities, during which time they were vulnerable to additional perturbations (Extended Data Fig. 5). For plants, widespread reproductive failures following heatwaves⁵⁹, such as those observed in lupines, could have long-term population consequences. While seedbanks can temporarily buffer such losses, losing entire seed cohorts during extreme climatic events may erode long-term seedbank persistence⁶⁰. Plants may also incur lagged mortality following heatwaves; our resampling of prickly rose the summer following the heatwave revealed dieback of main stems and associated basal resprouting in a quarter of plants. Second, long-term heatwave impacts may manifest through altered species interactions. Many of the impacted plants and sessile invertebrates in our study are habitat-forming species whose presence largely determines community diversity and composition⁶¹. Some of the impacted intertidal invertebrate and plant species are also important food resources^{62,63}; the decline in surf scoters 6 months after this heatwave may be linked to mass mortality of their intertidal invertebrate prey. Such indirect effects are becoming an increasingly documented impact of extreme heat⁶⁴.

Ecosystem-level impacts of the heatwave spanned immediate physiological responses and cross-seasonal disturbances that propagated well beyond the heatwave. Increases in temperature and VPD had variable effects on primary production during the heatwave itself, with GPP increasing in some areas and decreasing in others. On average, the region experienced a 10.9% decrease in GPP during the heatwave period, consistent with previous reports of GPP depression during extreme heat events in terrestrial⁶⁵ and marine ecosystems⁵¹. At high elevations, the extreme temperatures resulted in an unusually early and intense melt of snow and ice, leading to a late summer streamflow deficit, amplifying existing trends due to glacier retreat^{47,48}. Reduced streamflow can result in extremely high water temperatures in late summer, restructuring macroinvertebrate communities⁶⁶, enabling the spread of warm-adapted invasive species⁶⁷ and negatively impacting spawning salmon^{68,69}. Alpine meadow vegetation, including the lupines that suffered reproductive failure during the heatwave, also relies on water released by melting snow and ice throughout the summer⁷⁰. Increased wildfire activity in 2021 was triggered by high temperatures and widespread drought that dried flammable forest fuels⁷¹. Severe wildfires remove vegetation cover and reduce water infiltration, hastening soil erosion by removing vegetation, and increasing the likelihood of floods, debris flows and landslides⁷². Indeed, BC experienced historic flooding in the months following the heatwave, when intense rain fell in the watersheds encompassing the burn scars of several fires⁷³. Thus, the indirect and cascading ecosystem disturbances triggered by this extreme heatwave have far-reaching impacts beyond the species we documented.

To develop the best possible understanding of the ecological effects of this extreme but short-lived weather event, we compiled data from sources as disparate as remote sensing, regional streamflow databases, process-based plant ecophysiology models, long-term monitoring projects, camera traps, bio-acoustic monitors, citizen science surveys, repeated plot photographs, manipulative field experiments and opportunistic observations by scientists and the media. While this mixed-methods approach enabled us to quantify responses at a range of scales, from organisms to landscapes, we recognize that there were gaps in our synthesis, ranging from data sources and types

to taxonomic groups and response types. For example, only one contributing study was conducted in partnership with Indigenous peoples, with an Elder from the partnering Nation contributing knowledge and observations of species' responses from throughout their traditional territory. We acknowledge increasing calls to strengthen natural science and monitoring through meaningful collaborations with Indigenous peoples, not only as a means to better understand patterns of landscape and climate change by drawing on Indigenous place-based knowledges but also to uphold Indigenous rights and data sovereignty throughout their territories^{74–76}. Freshwater fishes and ecosystems were poorly represented despite intensive monitoring of salmonid stocks in the region because these efforts typically involve a single census in the late summer or fall, making it difficult to capture the impacts of discrete events. This highlights the limitations of relying on routine monitoring protocols, most of which were developed for different purposes than to capture biological responses to extreme weather events. Analogously, for mammals and birds, continuous data recorders detected (mostly muted) behavioural responses but not demographic ones, and the latter remain largely unknown for these species. More generally, a scientific challenge in assessing the consequences of unexpected, short-lived climatic extremes is to obtain adequate controls⁶. Here, we used spatial controls (exposed versus sheltered microclimates) and temporal controls (previous year(s) versus 2021), both of which showed net negative effects in our meta-analysis. These challenges should be considered for future studies of heatwave events.

With the frequency of extreme heatwaves forecasted to increase globally under climate change^{77,78}, it is critical that we gain improved understanding of the ecological effects of these events. We therefore propose three priorities in anticipation of and response to future heatwaves: (1) improved documentation of the modes of impact across species and ecosystems; (2) advanced understanding of the processes underlying ecological responses to heatwaves to improve ecological forecasting; and (3) well-considered deployment of management measures with assessment of their efficacy in an adaptive management framework. The capacity to achieve these goals hinges to a large extent on preparation well in advance of the next inevitable heatwave. Moving forward, we therefore encourage ecologists and environmental practitioners to build and strengthen networks, coordinating among academics and their institutions, government scientists and agencies, First Nations, non-profits and community scientists. Given limited resources, efforts should focus on the most vulnerable species, ecosystems and ecological processes as well as those of greatest concern and value for First Nations and other partners. Our results suggest, for example, that species with limited mobility, those that live in the intertidal zone and those sensitive to fire may be especially vulnerable. Recommended approaches to advance priorities 1 and 2 include enhancing the coverage and compatibility of targeted monitoring efforts for high-risk species and ecosystems⁷⁹; harnessing the power of growing open source ecological databases (for example, ref. 80) and artificial intelligence, while adhering to FAIR (Findable, Accessible, Interoperable and Reusable) principles for open data with CARE (Collective benefit, Authority to control, Responsibility and Ethics) principles for Indigenous data governance⁸¹; applying causal inferential statistical methods to observational heatwave datasets^{82,83} coupled with mechanistic laboratory and field experiments; and developing rapid response protocols that allow baseline sampling immediately in advance of forecasted heatwaves for the most vulnerable species (and life stages), habitats and locations. Focal questions to address key knowledge gaps include: How do differences in the magnitude, duration and timing of thermal exposure (for example, the effects of night-time heat) influence ecological impacts and recovery trajectories? How do shifts in community composition and species interactions, such as the loss of foundation species, influence longer-term biodiversity outcomes? How will processes fundamental to the biosphere, such as photosynthesis and carbon cycling, respond over short and longer timescales?

Finally, a number of management measures could be implemented to alleviate future heatwave impacts (priority 3). Many agricultural and aquaculture operations already use active strategies such as misting or spraying water over vulnerable fruit crops or beach-grown oysters, and new strategies could be developed. Landscape management could emphasize: (1) access to thermal refugia (for example, water sources, riparian buffers, corridors to cooler alpine habitats and urban green spaces); (2) connectivity between protected areas; and (3) fire management strategies including ecologically appropriate mechanical fuel treatments, prescribed burning and managed wildfire^{49,84–87}. In addition, for the most at-risk or impacted species, managers may consider greater interventions such as translocations, restoration or assisted evolution⁸⁸.

This study underscores the urgent need for detailed data to understand system baseline states, vulnerabilities and resilience to extreme events. This could be provided by coordinated cross-sectoral partnerships that can fund, plan and execute systematic, long-term biodiversity monitoring programs coupled with well-designed experiments and management interventions^{83,89,90}. The 2021 western North American heatwave underscores how critical the need is for such efforts and their resulting ecological insights, which will only increase this century as climate change intensifies.

Methods

Quantifying the 2021 heatwave

We defined the geographic region of the 2021 western North American heatwave by identifying the areas that experienced at least five consecutive days above the 90th percentile of their long-term maximum air temperature, as in Fischer and Schär⁹¹, for each corresponding calendar day of the most intense heatwave period, 25 June to 2 July 2021. To do so, we used Oak Ridge National Laboratory's Daymet Daily Surface Weather and Climatological Summaries, which provides gridded estimates of daily weather and climatology variables across the North American continent at a fine spatial resolution (1 km²) back to 1980⁹². Parameters in this dataset are generated using a network of ground-based observations as input data which is then interpolated and extrapolated through statistical modelling techniques. Considering the potential wide geographical area influenced by this heatwave, daily surface weather data for all of North America were directly downloaded.

Next, to illustrate the distinct forms of stress exposure during extreme heatwaves, we mapped air temperature (estimated at a height of 2 m above the land surface), land surface temperature (for example, soil, vegetation and asphalt) and VPD across the geographic extent of the heatwave, comparing historical and 2021 heatwave conditions. Air and land surface temperatures can decouple substantially from one another and hence pose different intensities of thermal stress during heatwaves. For example, high incoming solar radiation can heat the land surface substantially above the ambient air temperature, whereas high rates of evapotranspiration (for example, from soil or vegetation) may cool the land surface. These two metrics reflect—albeit imperfectly—different aspects of organismal thermal exposure, with the tissue temperature of a sessile organism, such as vegetation, likely to be more closely coupled to land surface temperature, whereas the tissue temperature of a motile organism, which can behaviourally thermoregulate, may be more closely coupled to air temperature. Moderate Resolution Imaging Spectroradiometer (MODIS) Land Surface Temperature daily data were retrieved at 1-km pixels by the generalized split-window algorithm and at 6-km grids by the algorithm (MOD11A1; version 6.1; from 2002 to present; <https://modis.gsfc.nasa.gov/data/dataproduct/mod11.php>). Daytime land surface temperatures for each day were measured around 13:00, and cloud-covered pixels were removed for further analysis. We calculated VPD from the Daymet daily average air temperature (T_{avg}) and vapour pressure (vp) measurements as the deviations between the estimated vapour pressure under the given thermal conditions and the actual amount of air moisture, as:

$$VPD = (0.61078 \times \exp[(17.269)(T_{avg})]) / (T_{avg} + 237.8) - vp,$$

following ref. 93, with greater VPD values indicating more severe drought conditions. We calculated historical average air temperature and VPD as the average of each variable over the available climatology (1980–2020) for the daytime period for the same days as the 2021 heatwave (day 176–183, that is, 25 June to 2 July) for all 1-km pixels in the heatwave-affected region. We then aggregated analogous averages for the heatwave days in 2021, as well as the maxima temperature and air moisture deficit (that is, VPD) experienced during the heatwave. Then, we calculated anomalies between averages (2021 heatwave averages and historical averages for the same period) and extremes (2021 heatwave maxima and historical averages) to reflect the general and the worst climatic conditions that each location was exposed to. Similarly, the average and extreme anomalies were calculated for surface temperature but using MODIS records from 2000 to 2020 as reference.

Finally, to highlight the temporal pattern of the 2021 heatwave, we aggregated the averages, 25th and 75th percentiles across variables for each calendar day using the 1980–2020 historical daily maximum and minimum temperature and estimated VPD. To identify the overall severity of the heatwave, climate anomalies during the 2021 heatwave from across its geographic extent were computed by subtracting 2021 daily temperature and VPD records from the averaged daily values of the 41-year reference period (that is, 1980–2020).

Conceptual thermal performance model

To illustrate the interacting effects of climate, climate variability and organismal physiology, we developed a conceptual model. We first simulated two example thermal performance curves (TPC) representing two hypothetical species which differ in their optimal performance temperature (T_{opt}) and performance breadth. These thermal performance curves were simulated using a modified Sharpe–Schoolfield model⁹⁴, with the following parameter values: for species 1, $E = 0.66$ eV, $E_D = 3.5$ eV, $T_{opt} = 25$ °C and for species 2, $E = 0.5$ eV, $E_D = 2$ eV, $T_{opt} = 33.5$ °C, where E is the apparent activation energy that governs the rate of increase in performance at suboptimal temperatures, E_D is the deactivation energy that characterizes the decrease in performance caused by heat-induced inactivation of enzymes or other biochemical components and T_{opt} is the optimal temperature. These parameter values were chosen to represent two contrasting hypothetical species with different thermal optima and performance breadths and fall within the empirical ranges of activation energies, deactivation energies and optimal temperatures reported across diverse taxa⁹⁵. Both models were normalized such that the maximum performance was equal to 1. Species performance distributions under historical average and heat dome conditions were estimated by multiplying the land surface temperature distributions during the historical average period and the heat dome period by each TPC. To illustrate the interaction between thermal optimum and thermal breadth on overall organismal performance, we performed this same analysis while varying T_{opt} and breadth through ranges of 15–35 °C and 22–50 °C, respectively. At each increment, the performance distribution was integrated and relative change in performance between historical and heatwave conditions was calculated to compute a heat map. This workflow used R packages *plantecophys* (vers. 1.4-6) and *rTPC* (vers. 1.0.4).

Species responses

We endeavoured to obtain broad documentation of the ecological effects of the 2021 heatwave across taxa and ecosystems. To initiate this effort, we convened a 'Heat Dome Symposium' and working group for ecologists in Western Canada who had collected relevant data during the event. We then contacted additional potential sources of information on the basis of our contacts and web searches, including Canadian and US academics; government scientists, eNGO and First Nations scientists, local botanical gardens and natural history societies,

specifically noting in our data requests that we were interested in all types of species responses (positive, neutral or negative) to the 2021 heatwave. We also searched ISI Web of Science and Google Scholar for relevant primary literature, using the terms ‘heatwave’ or ‘heat wave’ or ‘heat dome’ and ‘2021’, on 27 July 2022, 21 February 2023 and again on 17 July 2023; we also explored community science data sources (for example, Audubon’s e-bird) for relevant data.

In total, we compiled quantitative observations for 32 taxa of terrestrial insects, birds, mammals and plants, as well as intertidal invertebrates, seagrass and seaweeds across a ~600,000-km² range throughout the area of Canada affected by the 2021 heatwave that fit the criteria for inclusion in the meta-analysis (Extended Data Fig. 6, Extended Data Table 2 and Supplementary Methods). Studies had to (1) provide quantitative measure(s) of an ecological response for one or more species to the 2021 western North American heatwave, including the sample size and some measure of variance and (2) compare the response(s) to either a temporal control (that is, before versus during and/or after the heatwave) or a spatial control (that is, with varying exposure to the heatwave). Datasets that met these criteria and were included in our meta-analysis came from independent studies, monitoring programs or experiments running at the time of the 2021 heatwave (Supplementary Methods). We could not include any of the relevant studies from literature searches either because they did not include a control or did not include an estimate of variance. Response variables, which included measures specific to the objectives and methods of each individual study (Extended Data Table 2), were categorized as persistence (that is, changes in abundance or survival), behaviour (for example, changes in activity level), damage (for example, leaf scorch), growth or mortality. We also categorized each study organism as sessile or motile, terrestrial or marine by its major taxonomic group and by when the timing of the measurements occurred (for example, during or after the heatwave) to include as predictor variables in our multilevel meta-analytic model (Extended Data Table 2).

For each dataset that met our criteria for inclusion, we calculated the mean, s.d. and sample size of the response for the control (that is, either the time period before the 2021 heatwave or a spatial control) and the heatwave exposure. We then quantified the effects of the 2021 heatwave across species and responses using log-transformed response ratios (lnRR) as effect sizes, which compare the ratio of means between two groups⁹⁶. For temporal comparisons, we defined lnRR to be the ratio of means between responses to the 2021 heatwave either after (later in 2021 or the following year, 2022) or during (25 June to 2 July 2021) and the response before the 2021 heatwave (before 25 June 2021). For spatial comparisons, we defined lnRR to be the ratio of means between more exposed habitats during the heatwave (for example, no shade) and more buffered habitats (for example, shade) (see Extended Data Table 2 for the control level specified in each study). To compare the effects of the 2021 heatwave on responses across organisms and ecosystems, we then analysed the lnRRs and corresponding sampling variances in a multilevel meta-analytic model using the R package metafor (v 4.3-2)⁹⁶. With the nature of collating many disparate studies into a single model, we maintained the importance of a random effect for study throughout our model analysis and included a variance covariance matrix to account for shared controls within each study. We also specified restricted maximum likelihood as our variance estimation method owing to our small sample size and additional terms to adjust our estimation of degrees of freedom (d.f.)^{96,97}. We first ran an overall model with a random effect for study but no moderator. We then tested a full model with fixed effects of: broad taxonomic group, movement, timing and response type. We then calculated variance inflation factors (VIFs) for each predictor in this full model to assess collinearity among predictor variables, with a VIF threshold of 3 to indicate moderate collinearity⁹⁸. If any predictor exceeded a VIF threshold of 3, we sequentially removed predictors until all VIFs were below the threshold of 3 (refs. 98,99). This resulted in taxonomy and response

type being omitted from the model. Once these collinear predictors were omitted from the model, we applied a cluster-robust variance estimation method to calculate adjusted estimates on the basis of a variance–covariance matrix of the model coefficient estimates via a sandwich type estimator⁹⁷. We inferred strong support for a given effect if the model estimate was greater than 0.8, moderate support if the model estimate was 0.5 and weak support if the model estimate was less than 0.2 (ref. 100) and inferred statistical significance on the basis of whether CIs overlapped zero.

In addition to the meta-analysis, we also collated records of ecological responses to the 2021 western North American heatwave from further online searches, recognizing that given the lack of forewarning about the event, many such observations were made outside of formal studies. This included online media reports of wildlife observations made during the heatwave, from wildlife rehabilitation centres, public photos and other sources, as well as casual observations made by ecologists during the event. We also included those studies from the literature searches that were relevant, in that they described an ecological response to the 2021 western North American heatwave, but did not meet the criteria for inclusion in the meta-analysis. To ensure reliability, we included only observations for which the taxon, location, timing and linkage to the heatwave could be clearly verified, and we excluded reports that lacked these details or presented ambiguous attribution.

Ecosystem function change

To estimate changes during the 2021 heatwave to a key terrestrial ecosystem flux, GPP, we used a coupled model of photosynthesis and stomatal conductance based on the Farquhar–von Caemmerer–Berry model (FvCB)^{101–103}, a biochemical model widely used in land surface modelling. The FvCB model predicts leaf-level area-normalized gross photosynthetic rate as the minimum of three biochemical processes: Rubisco carboxylation, RuBP regeneration and triose phosphate utilization. We parameterized the FvCB model using the Community Land Model 5 (CLM5)¹⁰⁴; average parameters for the needleleaf evergreen tree plant functional type, which is the dominant vegetation type in the area of the 2021 heatwave, and photosynthesis was simulated in each grid cell using the climate data described above (see ‘Quantifying the 2021 heatwave’ section). We ran the FvCB model on each grid cell within the heatwave area, first using the historical land surface temperature, mean 10-day air temperature (to account for temperature acclimation)¹⁰⁴ and VPD values, then using the corresponding climate data from 2021. Percentage change in GPP for each grid cell was then computed as $\delta = (GPP_{2021} - GPP_{hist}) / GPP_{hist}$, where δ is the percent change in modelled photosynthetic rate during the 2021 heatwave, relative to the historical average, GPP_{2021} is the modelled mean GPP during the period from 25 June to 2 July 2021 and GPP_{hist} is the modelled mean GPP during 25 June to 2 July historical average conditions. This approach enabled us to estimate the relative change in GPP within each grid cell but it does not account for variation in leaf area index or other structural or functional type variation that may occur between grid cells in the area of the heatwave, and thus we were unable to estimate the total change in GPP over the entire region of the heatwave. We estimated mean thermal performance of GPP across the study area by regressing land surface temperatures against VPD using the gridded climate data from the historical average period and the 2021 heatwave. We then ran the FvCB model using leaf temperatures ranging from 0 to 50 °C and corresponding VPD values predicted from the regression to estimate thermal optima during the historical period and the period of the 2021 heatwave. This workflow used R packages plantecophys (vers. 1.4-6) and rTPC (vers. 1.0.4).

Wildfires and streamflow

To map wildfire activity, we used remote sensing imagery from the MODIS Level 3 8-Day composite active fire product (MOD14A2; Giglio and Justice, 2015). This product is a gridded 1-km resolution composite

of the individual fire pixel classes observed over 8-day periods. In situations where a pixel varies in fire class within those 8 days, the maximum value is taken. We extracted data for the period of 1 May to 30 September for every year since 2000, projected the data to an Albers Equal Area projection to preserve area and maintain accurate area measurements and clipped to the geographic area exposed to the 2021 heatwave (as defined above). For each 8-day period in each year, we summed the number of pixels (number of $1\text{ km} \times 1\text{ km}$ grid cells) within the area covered by the 2021 heatwave that were classified as 'medium' or 'high' confidence for fire activity. We then calculated the mean (± 1 standard error) of the past 5 years as well as the past 20 years. The 8-day intervals were offset by 1 day in leap years (2000, 2004, 2008, 2012, 2016 and 2020). However, as fire area was measured over rolling intervals (rather than single days), we assumed that this offset would not significantly bias our results. Therefore, for the purposes of calculating mean fire activity at each 8-day interval for the past 5 years and 20 years, the same interval period was assigned for both leap years and non-leap years. To determine whether fire activity in 2021 was significantly different from historical levels, we conducted one-sample two-tailed *t*-tests comparing fire activity for 2021 with fire activity from the sample set consisting of the previous 20 years. We conducted each test for fire activity levels measured during the week of the heat dome event (25 June to 2 July) and the week following the heat dome (3 July to 9 July).

We quantified impacts of the 2021 heatwave on streamflows by comparing 2021 streamflows with those from the previous two decades and evaluated if local temperature, glaciers and snow cover modulated the impacts. We downloaded streamflow information for the period 2000–2021 for all 183 Water Survey of Canada stations within the area affected by the 2021 heatwave (as previously defined) with complete information on stream flow, catchment area, snow cover, glacier extent and temperature change; these criteria necessarily restricted the stations to high altitude sites. To have robust comparisons between 2021 and previous years, we also only included stations for which streamflow data were available for at least 50% of the days in each year and at least 80% of the years in the reference period of 2000–2020. Streamflow data were downloaded from the Environment and Climate Change Canada Historical Hydrometric Data (HYDAT) database (https://wateroffice.ec.gc.ca/mainmenu/historical_data_index_e.html) using the tidyhydat R package (v 0.7.2); heatwave areas in the USA were not included, as comparable streamflow and catchment data were not readily available. These data were last accessed on 26 November 2025, with the HYDAT data versioned 14 October 2025. We then extracted catchment shape files for each streamflow station from the National Hydrometric Network Basin Polygon dataset (<https://open.canada.ca/data/en/dataset> with record ID no. 0c121878-ac23-46f5-95df-eb9960753375). We also downloaded glacier polygons from the Randolph Glacier Inventory v6.0 database (RGI Consortium, 2017) to calculate the percent of catchment covered by glaciers. For each catchment, we extracted the amount of snow cover from MODIS/Terra Snow Cover Daily L3 Global 500 m SIN Grid, version 61 (MOD10A1) (DOI: 10.5067/MODIS/MOD10_L2.061) for 24 June 2021 with a 2-day window on either side. We also calculated the amount of snow cover within the glaciated area in the catchment, as glaciers may reduce rates of snow melt. Finally, we calculated the change in temperature at each streamflow station as (1) deviation of the mean daily maximum air temperature in the heatwave period (25 June to 2 July 2021) from the previous week (17–24 June 2021) and (2) the temperature anomaly of mean daily maximum surface temperatures in the period 25 June to 2 July 2021 from that of the previous 40 years. All temperature data were extracted from the Daymet database (<https://daymet.ornl.gov/citations>).

Our response variables were the ratio of 2021 streamflow to the mean streamflow over 2000–2020, calculated both during the heatwave period as well as in late summer (1–31 August) to capture both immediate and seasonal effects. We first used one-sided *t*-tests to

analyse whether these ratios on average differed from one. We then used linear models to examine the effect of snow, ice and temperature on spatial variance in these ratios. Specifically, we used a model selection approach with the MuMIn R package (vers. 1.48.11) to choose the most parsimonious models describing the relationship between the streamflow ratios and maximum temperature deviations or surface air anomalies, percentage catchment coverage by snow cover and percent catchment coverage by glaciers (Extended Data Table 4). We ran models both weighted and unweighted by data quality to ensure that data quality did not influence results, and we transformed and scaled variables to ensure normality of residuals. To visualize the model results, which often involved complex three-way interactions, we collapsed the snow $\times$ glacier interaction into three categories on the basis of the medians of glacier cover and snow cover in our database: (1) 'glacier, high snow' had $>0.5\%$ glacier cover and $>25\%$ snow cover, (2) 'glacier, low snow' had $>0.5\%$ glacier cover and $<25\%$ snow cover and (3) 'no glacier, low snow' had $<0.5\%$ glacier cover and $<25\%$ snow cover. We did not include a possible fourth category of 'no glacier, high snow' ($<0.5\%$ glacier cover and $>25\%$ snow cover), as this combination was only represented by a single stream in the database; that is, it is very unusual for the conditions that lead to high snow cover in late June to not also lead to glacier formation.

Software

Unless otherwise stated, all analysis were completed using R 4.4.2.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The ecological datasets owned by the authors and used in the analyses herein are publicly available via Zenodo at <https://doi.org/10.5281/zenodo.17833420> (ref. 105). All other datasets are open source.

Code availability

The R code used in the analyses herein is publicly available via Zenodo at <https://doi.org/10.5281/zenodo.17833420> (ref. 105).

References

- Stillman, J. H. Heat waves, the new normal: summertime temperature extremes will impact animals, ecosystems, and human communities. *Physiology* **34**, 86–100 (2019).
- IPCC. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (eds Masson-Delmotte, V. et al.) (Cambridge Univ. Press, 2021).
- Sergio, F., Blas, J. & Hiraldo, F. Animal responses to natural disturbance and climate extremes: a review. *Glob. Planet. Chang.* **161**, 28–40 (2018).
- Neilson, E. W. et al. There's a storm a-coming: ecological resilience and resistance to extreme weather events. *Ecol. Evol.* **10**, 12147–12156 (2020).
- IPCC. *Climate Change 2022: Impacts, Adaptation, and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (eds Pörtner, H.-O. et al.) (Cambridge Univ. Press, 2022).
- Evans, M. E. K., Hu, J. & Michaletz, S. T. Scaling plant responses to heat: from molecules to the biosphere. *Science* **388**, 1167–1173 (2025).
- Sibley, A. et al. Extreme heatwave causes immediate, widespread mortality of forest canopy foliage, highlighting modes of forest sensitivity to extreme heat. *Glob. Chang. Biol.* **31**, e70571 (2025).
- Zscheischler, J. et al. Future climate risk from compound events. *Nat. Clim. Chang.* **8**, 469–477 (2018).

9. AghaKouchak, A. et al. Climate extremes and compound hazards in a warming world. *Annu. Rev. Earth Planet. Sci.* **48**, 519–548 (2020).
10. Meisner, A., De Deyn, G. B., De Boer, W. & Van Der Putten, W. H. Soil biotic legacy effects of extreme weather events influence plant invasiveness. *Proc. Natl Acad. Sci. USA* **110**, 9835–9838 (2013).
11. Nadeau, C. P., Urban, M. C. & Bridle, J. R. Climates past, present, and yet-to-come shape climate change vulnerabilities. *Trends Ecol. Evol.* **32**, 786–800 (2017).
12. Williams, S. E., Shoo, L. P., Isaac, J. L., Hoffmann, A. A. & Langham, G. Towards an integrated framework for assessing the vulnerability of species to climate change. *PLoS Biol.* **6**, e325 (2008).
13. Cunningham, S. J., Gardner, J. L. & Martin, R. O. Opportunity costs and the response of birds and mammals to climate warming. *Front. Ecol. Environ.* **19**, 300–307 (2021).
14. Scherrer, D. & Körner, C. Infra-red thermometry of alpine landscapes challenges climatic warming projections. *Glob. Chang. Biol.* **16**, 2602–2613 (2010).
15. Starko, S. et al. Microclimate predicts kelp forest extinction in the face of direct and indirect marine heatwave effects. *Ecol. Appl.* **32**, e2673 (2022).
16. Blonder, B., Escobar, S., Kapás, R. E. & Michaletz, S. T. Low predictability of energy balance traits and leaf temperature metrics in desert, montane and alpine plant communities. *Funct. Ecol.* **34**, 1882–1897 (2020).
17. Leuzinger, S. & Körner, C. Tree species diversity affects canopy leaf temperatures in a mature temperate forest. *Agric. For. Meteorol.* **146**, 29–37 (2007).
18. Porter, W. P. & Kearney, M. Size, shape, and the thermal niche of endotherms. *Proc. Natl Acad. Sci. USA* **106**, 19666–19672 (2009).
19. Gates, D. M. *Biophysical Ecology* (Springer-Verlag, 1980).
20. Michaletz, S. T. et al. The energetic and carbon economic origins of leaf thermoregulation. *Nat. Plants* **2**, 16129 (2016).
21. De Frenne, P. et al. Global buffering of temperatures under forest canopies. *Nat. Ecol. Evol.* **3**, 744–749 (2019).
22. Garen, J. C. et al. Canopy-top measurements do not accurately quantify canopy-scale leaf thermoregulation. *Proc. Natl Acad. Sci. USA* **120**, e2301914120 (2023).
23. Terrien, J. Behavioral thermoregulation in mammals: a review. *Front. Biosci.* **16**, 1428 (2011).
24. Bicego, K. C., Barros, R. C. H. & Branco, L. G. S. Physiology of temperature regulation: comparative aspects. *Comp. Biochem. Physiol. A* **147**, 616–639 (2007).
25. Menzies, A. K. et al. Body temperature, heart rate, and activity patterns of two boreal homeotherms in winter: homeostasis, allostasis, and ecological coexistence. *Funct. Ecol.* **34**, 2292–2301 (2020).
26. Gilbert, N. A., Stenglein, J. L., Van Deelen, T. R., Townsend, P. A. & Zuckerberg, B. Behavioral flexibility facilitates the use of spatial and temporal refugia during variable winter weather. *Behav. Ecol.* **33**, 446–454 (2022).
27. Hetem, R. S., Fuller, A., Maloney, S. K. & Mitchell, D. Responses of large mammals to climate change. *Temperature* **1**, 115–127 (2014).
28. Verzuh, T. L. et al. Behavioural responses of a large, heat-sensitive mammal to climatic variation at multiple spatial scales. *J. Anim. Ecol.* **92**, 619–634 (2023).
29. Beever, E. A. et al. Behavioral flexibility as a mechanism for coping with climate change. *Front. Ecol. Environ.* **15**, 299–308 (2017).
30. Thompson, V. et al. The 2021 western North America heat wave among the most extreme events ever recorded globally. *Sci. Adv.* **8**, eabm6860 (2022).
31. Canada's top 10 weather stories of 2021. ECCC <https://www.canada.ca/en/environment-climate-change/services/top-ten-weather-stories/2021.html> (21 December 2021).
32. White, R. H. et al. The unprecedented Pacific Northwest heatwave of June 2021. *Nat. Commun.* **14**, 727 (2023).
33. Philip, S. Y. et al. Rapid attribution analysis of the extraordinary heat wave on the Pacific coast of the US and Canada in June 2021. *Earth Syst. Dynam.* **13**, 1689–1713 (2022).
34. Bartusek, S., Kornhuber, K. & Ting, M. 2021 North American heatwave amplified by climate change-driven nonlinear interactions. *Nat. Clim. Change* **12**, 1143–1150 (2022).
35. Henderson, S. B., McLean, K. E., Lee, M. J. & Kosatsky, T. Analysis of community deaths during the catastrophic 2021 heat dome: early evidence to inform the public health response during subsequent events in greater Vancouver, Canada. *Environ. Epidemiol.* **6**, e189 (2022).
36. *Extreme Heat and Human Mortality: A Review of Heat-Related Deaths in B.C. in Summer 2021* (The BC Coroners Service, 2022).
37. Tetzlaff, E. J., Goulet, N., Gorman, M., Ioannou, L. G. & Kenny, G. P. Working under the 2021 heat dome: a content analysis of occupational impacts mentioned in the Canadian media. *Healthcare* **11**, 2423 (2023).
38. Harley, C. D. G. Climate change, keystone predation, and biodiversity loss. *Science* **334**, 1124–1127 (2011).
39. Michaletz, S. T. & Garen, J. C. Hotter is not (always) better: embracing unimodal scaling of biological rates with temperature. *Ecol. Lett.* **27**, e14381 (2024).
40. Sloane, S. A., Gordon, A. & Connelly, I. D. Bushtit (*Psaltirparus minimus*) nestling mortality associated with unprecedented June 2021 heatwave in Portland, Oregon. *Wilson J. Ornithol.* **134**, 155–162 (2022).
41. Neal, E., Huang, C. S. Y. & Nakamura, N. The 2021 Pacific Northwest heat wave and associated blocking: meteorology and the role of an upstream cyclone as a diabatic source of wave activity. *Geophys. Res. Lett.* **49**, e2021GL097699 (2022).
42. Johnson, E. A. & Wowchuk, D. R. Wildfires in the southern Canadian Rocky Mountains and their relationship to mid-tropospheric anomalies. *Can. J. For. Res.* **23**, 1213–1222 (1993).
43. Macias Fauria, M., Michaletz, S. T. & Johnson, E. A. Predicting climate change effects on wildfires requires linking processes across scales. *WIREs Clim. Change* **2**, 99–112 (2011).
44. 2021 Wildfire season summary. Province of British Columbia <https://www2.gov.bc.ca/gov/content/safety/wildfire-status/about-bcws/wildfire-history/wildfire-season-summary> (2021).
45. Littell, J. S., Peterson, D. L., Riley, K. L., Liu, Y. & Luce, C. H. A review of the relationships between drought and forest fire in the United States. *Glob. Chang. Biol.* **22**, 2353–2369 (2016).
46. Jain, P., Sharma, A. R., Acuna, D. C., Abatzoglou, J. T. & Flannigan, M. Record-breaking fire weather in North America in 2021 was initiated by the Pacific Northwest heat dome. *Commun. Earth Environ.* **5**, 1–10 (2024).
47. Bonsal, B. et al. Western Canadian freshwater availability: current and future vulnerabilities. *Environ. Rev.* **28**, 528–545 (2020).
48. Stahl, K. & Moore, R. D. Influence of watershed glacier coverage on summer streamflow in British Columbia, Canada. *Water Resour. Res.* **42**, W06201 (2006).
49. Maxwell, S. L. et al. Conservation implications of ecological responses to extreme weather and climate events. *Divers. Distrib.* **25**, 613–625 (2019).
50. Vasseur, D. A. et al. Increased temperature variation poses a greater risk to species than climate warming. *Proc. Biol. Sci.* **281**, 20132612 (2014).
51. Smale, D. A. et al. Marine heatwaves threaten global biodiversity and the provision of ecosystem services. *Nat. Clim. Change* **9**, 306–312 (2019).

52. Smith, K. E. et al. Global impacts of marine heatwaves on coastal foundation species. *Nat. Commun.* **15**, 5052 (2024).
53. Ruthrof, K. X. et al. Subcontinental heat wave triggers terrestrial and marine, multi-taxa responses. *Sci. Rep.* **8**, 13094 (2018).
54. Bonoan, R. E., Goldman, R. R., Wong, P. Y. & Starks, P. T. Vasculature of the hive: heat dissipation in the honey bee (*Apis mellifera*) hive. *Naturwissenschaften* **101**, 459–465 (2014).
55. Meikle, W. G., Barg, A. & Weiss, M. Honey bee colonies maintain CO₂ and temperature regimes in spite of change in hive ventilation characteristics. *Apidologie* **53**, 51 (2022).
56. Alston, J. M., Joyce, M. J., Merkle, J. A. & Moen, R. A. Temperature shapes movement and habitat selection by a heat-sensitive ungulate. *Landsc. Ecol.* **35**, 1961–1973 (2020).
57. Veldhuis, M. P. et al. Predation risk constrains herbivores' adaptive capacity to warming. *Nat. Ecol. Evol.* **4**, 1069–1074 (2020).
58. Wingfield, J. C. et al. How birds cope physiologically and behaviourally with extreme climatic events. *Philos. Trans. R. Soc. Lond. B* **372**, 20160140 (2017).
59. Breshears, D. D. et al. Underappreciated plant vulnerabilities to heat waves. *New Phytol.* **231**, 32–39 (2021).
60. Plue, J. et al. Buffering effects of soil seed banks on plant community composition in response to land use and climate. *Global Ecol. Biogeogr.* **30**, 128–139 (2021).
61. Hesketh, A. V. & Harley, C. D. G. Extreme heatwave drives topography-dependent patterns of mortality in a bed-forming intertidal barnacle, with implications for associated community structure. *Glob. Change Biol.* **29**, 165–178 (2023).
62. Visscher, D. R., Merrill, E. H., Fortin, D. & Frair, J. L. Estimating woody browse availability for ungulates at increasing snow depths. *For. Ecol. Manag.* **222**, 348–354 (2006).
63. Goodwin, K. *Herbivory and Climate as Drivers of Plant Population and Range Dynamics*. PhD thesis, Univ. British Columbia (2025).
64. Starko, S. et al. in *Oceanography and Marine Biology: An Annual Review* (eds Russell, B. D. & Todd, P. A.) 42–96 (CRC Press, 2025).
65. Ciais, P. et al. Europe-wide reduction in primary productivity caused by the heat and drought in 2003. *Nature* **437**, 529–533 (2005).
66. Jourdan, J. et al. Effects of changing climate on European stream invertebrate communities: a long-term data analysis. *Sci. Total Environ.* **621**, 588–599 (2018).
67. Daufresne, M., Bady, P. & Fruget, J.-F. Impacts of global changes and extreme hydroclimatic events on macroinvertebrate community structures in the French Rhône river. *Oecologia* **151**, 544–559 (2007).
68. Pelto, M. S., Dryak, M., Pelto, J., Matthews, T. & Perry, L. B. Contribution of glacier runoff during heat waves in the Nooksack River Basin USA. *Water* **14**, 1145 (2022).
69. Westley, P. A. H. Documentation of en route mortality of summer chum salmon in the Koyukuk River, Alaska and its potential linkage to the heatwave of 2019. *Ecol. Evol.* **10**, 10296–10304 (2020).
70. Winkler, D. E., Butz, R. J., Germino, M. J., Reinhardt, K. & Kueppers, L. M. Snowmelt timing regulates community composition, phenology, and physiological performance of alpine plants. *Front. Plant Sci.* **9**, 1140 (2018).
71. Parisien, M. A. et al. Abrupt, climate-induced increase in wildfires in British Columbia since the mid-2000s. *Nat. Commun. Earth Environ.* **4**, 309 (2023).
72. Rengers, F. K. et al. Landslides after wildfire: initiation, magnitude, and mobility. *Landslides* **17**, 2631–2641 (2020).
73. Logie, G. Wildfires and flood damage. *SparkGeo* sparkgeo.com/blog/wildfires-and-flood-damage/ (2022).
74. Ignace, L. et al. Researchers' responsibility to uphold Indigenous rights. *Science* **381**, 129–131 (2023).
75. Wong, C., Ballegooyen, K., Ignace, L., Johnson, M. J. (G. ùdia) & Swanson, H. Towards reconciliation: 10 calls to action to natural scientists working in Canada. *Facets* **5**, 769–783 (2020).
76. Jessen, T. D. et al. Indigenous peoples as sentinels of change in human-wildlife relationships: conservation status of mountain goats in Kitasoo Xai'xais territory and beyond. *Conserv. Sci. Pract.* **4**, e12662 (2022).
77. Meehl, G. A. & Tebaldi, C. More intense, more frequent, and longer lasting heat waves in the 21st century. *Science* **305**, 994–997 (2004).
78. Perkins-Kirkpatrick, S. E. & Lewis, S. C. Increasing trends in regional heatwaves. *Nat. Commun.* **11**, 3357 (2020).
79. Burton, A. C. et al. A framework for adaptive monitoring of the cumulative effects of human footprint on biodiversity. *Environ. Monit. Assess.* **186**, 3605–3617 (2014).
80. Kays, R. et al. The Movebank system for studying global animal movement and demography. *Methods Ecol. Evol.* **13**, 419–431 (2022).
81. Carroll, S. R., Herczog, E., Hudson, M., Russell, K. & Stall, S. Operationalizing the CARE and FAIR principles for Indigenous data futures. *Sci. Data* **8**, 108 (2021).
82. Arif, S. & MacNeil, M. A. Applying the structural causal model framework for observational causal inference in ecology. *Ecol. Monogr.* **93**, e1554 (2023).
83. Gonzalez, A., Chase, J. M. & O'Connor, M. I. A framework for the detection and attribution of biodiversity change. *Philos. Trans. R. Soc. Lond. B* **378**, 20220182 (2023).
84. Hylander, K., Greiser, C., Christiansen, D. M. & Koelemeijer, I. A. Climate adaptation of biodiversity conservation in managed forest landscapes. *Conserv. Biol.* **36**, e13847 (2022).
85. Hessburg, P. F. et al. Restoring fire-prone Inland Pacific landscapes: seven core principles. *Landsc. Ecol.* **30**, 1805–1835 (2015).
86. Daniels, L. D. et al. The 2023 wildfires in British Columbia, Canada: impacts, drivers, and transformations to coexist with wildfire. *Can. J. For. Res.* **55**, 1–18 (2025).
87. Prichard, S. J. et al. Adapting western North American forests to climate change and wildfires: 10 common questions. *Ecol. Appl.* **31**, e02433 (2021).
88. Ainsworth, T. D., Hurd, C. L., Gates, R. D. & Boyd, P. W. How do we overcome abrupt degradation of marine ecosystems and meet the challenge of heat waves and climate extremes?. *Glob. Change Biol.* **26**, 343–354 (2020).
89. Gonzalez, A. et al. A global biodiversity observing system to unite monitoring and guide action. *Nat. Ecol. Evol.* **7**, 1947–1952 (2023).
90. Gonzalez, A. et al. A Biodiversity Observation Network to support conservation action and mainstream knowledge in Canada. *Facets* **10**, 1–19 (2025).
91. Fischer, E. & Schär, C. Consistent geographical patterns of changes in high-impact European heatwaves. *Nat. Geosci.* **3**, 398–403 (2010).
92. Daymet: daily surface weather data on a 1-km grid for North America. DAAC https://daac.ornl.gov/DAYMET/guides/Daymet_Daily_V4.html (2024).
93. Murray, F. W. On the computation of saturation vapor pressure. *J. Appl. Meteorol.* **6**, 203–204 (1966).
94. Kontopoulou, D.-G., Smith, T. P., Barraclough, T. G. & Pawar, S. Adaptive evolution shapes the present-day distribution of the thermal sensitivity of population growth rate. *PLoS Biol.* **18**, e3000894 (2020).
95. Dell, A. I., Pawar, S. & Savage, V. M. Systematic variation in the temperature dependence of physiological and ecological traits. *Proc. Natl Acad. Sci. USA* **108**, 10591–10596 (2011).
96. Viechtbauer, W. Conducting meta-analyses in R with the metafor Package. *J. Stat. Soft.* **36**, 1–48 (2010).

97. Viechtbauer, W. Cluster-robust tests and confidence intervals for 'rma' objects. *metafor* 4.9-10 <https://www.viechtb.github.io/metafor/reference/robust.html> (2025).
98. Zuur, A. F., Ieno, E. N. & Elphick, C. S. A protocol for data exploration to avoid common statistical problems. *Methods Ecol. Evol.* **1**, 3–14 (2010).
99. Harrison, X. A. et al. A brief introduction to mixed effects modelling and multi-model inference in ecology. *PeerJ* **6**, e4794 (2018).
100. Sullivan, G. M. & Feinn, R. Using effect size—or why the P value is not enough. *J. Grad. Med. Educ.* **4**, 279–282 (2012).
101. Farquhar, G. D., Von Caemmerer, S. & Berry, J. A. A biochemical model of photosynthetic CO₂ assimilation in leaves of C3 species. *Planta* **149**, 78–90 (1980).
102. Medlyn, B. E. et al. Reconciling the optimal and empirical approaches to modelling stomatal conductance. *Glob. Change Biol.* **17**, 2134–2144 (2011).
103. Duursma, R. A. Plantecophys - An R package for analysing and modelling leaf gas exchange data. *PLoS ONE* **10**, e0143346 (2015).
104. Lawrence, D. M. et al. The Community Land Model version 5: description of new features, benchmarking, and impact of forcing uncertainty. *J. Adv. Model Earth Syst.* **11**, 4245–4287 (2019).
105. Baum, J. K. et al. Data and Code for article - Widespread ecological responses and cascading effects of the 2021 western North American heatwave v1.0.1. *Zenodo* <https://doi.org/10.5281/zenodo.17833420> (2025).
106. McAfee, A. et al. Vulnerability of honey bee queens to heat-induced loss of fertility. *Nat. Sustain.* **3**, 367–376 (2020).
107. Groh, C., Tautz, J. & Rössler, W. Synaptic organization in the adult honey bee brain is influenced by brood-temperature control during pupal development. *Proc. Natl Acad. Sci. USA* **101**, 4268–4273 (2004).
108. Dole, B. Baby hawks in Oregon bailed from nests amid heatwave: 'They had no choice'. *The Oregonian* <https://www.oregonlive.com/environment/2021/07/baby-hawks-in-oregon-bailed-from-nests-amid-heatwave-they-had-no-choice.html> (9 July 2021).
109. Labbé, S. June heat wave affected birds in B.C., Washington and Oregon. *Pique News Magazine* <https://www.piquenewsmagazine.com/must-reads/birds-jumped-out-of-their-nests-to-escape-the-june-heat-wave-4183127> (30 July 2021).
110. Record heat kills more than 100 birds at nesting spot near Duwamish River. *West Seattle Blog* <https://westseattleblog.com/2021/06/wildlife-record-heat-kills-more-than-100-birds-at-nesting-spot-near-duwamish-river/> (30 June 2021).
111. Thompson, J. Why heat waves amplified by climate change are a big problem for baby birds. *Audubon* <https://www.audubon.org/news/why-heat-waves-amplified-climate-change-are-big-problem-baby-birds> (2021).
112. Srivastava, D. S. et al. Insects in the city: determinants of a contained aquatic microecosystem across an urbanized landscape. *Ecology* **105**, e4204 (2024).
113. Labbé, S. Shellfish warning issued after spike in illnesses linked to B.C. heat wave. *Times Colonist* <https://www.timescolonist.com/local-news/shellfish-warning-issued-after-spike-in-illnesses-linked-to-bc-heat-wave-4690929> (26 July 2021).
114. Lim, E. G., Reed, K. J., Campbell, J. A. & Côté, I. M. Localised mass mortality of giant California sea cucumbers in Western Canada. *Mar. Biol.* **170**, 86 (2023).
115. Aoki, L. R. et al. UAV high-resolution imaging and disease surveys combine to quantify climate-related decline in seagrass meadows. *Oceanography* **36**, 38–39 (2023).
116. Lundeberg, S. Pacific Northwest heat dome tree damage more about temperature than drought, scientists say. *Oregon State University News* <https://news.oregonstate.edu/news/pacific-northwest-heat-dome-tree-damage-more-about-temperature-drought-scientists-say> (3 February 2023).
117. Gomez, M. B.C. heat wave 'cooks' fruit crops on the branch in sweltering Okanagan and Fraser valleys. *CBC* <https://www.cbc.ca/news/canada/british-columbia/heat-fruit-crops-okanagan-fraser-valley-1.6092155> (6 July 2021).

Acknowledgements

This publication was the result of a working group held at the University of British Columbia (UBC) in May 2022. The working group was organized by the Canadian Institute of Ecology and Evolution and funded by the UBC's Biodiversity Research Centre through a Grant for Catalyzing Research Clusters. We thank the Beaty Biodiversity Museum and the Peter Wall Institute for Advanced Studies for use of their facilities to host the working group and associated public symposium. This study was also supported during the data collection and synthesis phases by the Natural Sciences and Engineering Research Council of Canada (NSERC), including through Discovery Grants to J.K.B. (grant nos. RGPIN-06669-2018 and RGPIN-2024-04770 and Discovery Grant Accelerator no. RGPAS-522477-2018), C.D.G.H. (grant no. RGPIN-2016-05441), D.S.S. (grant nos. RGPIN-2016-03813 and RGPIN-2022-04555), A.L.A. (grant no. RGPIN-2019-010335) and M.M. (grant no. DGEGR-2022-00342) and scholarships and fellowships to B.T. and S.S. Data collection by H.E., A.-L.M.G. and G.S.-B. was supported by the Tula Foundation; data collection by A.C.B., T.G., C.S. and K.T.-M. was supported by the British Columbia Ministries of Water, Land, and Resource Stewardship (grant no. TP22AR004) and Environment and Climate Change Strategy (grant no. TP22JHQ018); data collection by S.D.H., L.D.D. and B.W. was supported by the Intact Foundation and the UBC Future Forests Fellowship; and data collection by M.M. was also supported by UBC's Campus as a Living Laboratory Fund, while that of P.K.A., M.F., Y.U. and J.T. was provided by a Canadian Agriculture Partnership Agri-Science Project (grant no. 3175), a partnership between Agriculture and Agri-Food Canada and the Lower Mainland Horticultural Improvement Association.

Author contributions

D.S.S., J.K.B., M.T., S.T.M., K.J.A.G., S.E., M.A.S., C.G.C., C.D.G.H., A.C.B., Z.S., J.L. and J.C.G. collectively conceived and designed the tests of heat dome impacts. J.K.B., M.T., J.C.G., Z.S., S.E., K.J.A.G., J.L., M.T., S.T.M., A.C.B., C.D.G.H. and D.S.S. obtained and curated online or primary datasets. Z.S., J.C.G., M.A.S., S.E., D.S.S., J.L., S.T.M., C.D.G.H., C.G.C., A.C.B., K.J.A.G. and M.T. analysed and/or visualized the data, with input on all figures from J.K.B., C.D.G.H., A.C.B., K.J.A.G., S.E. and the 2021 Western North American Heat Dome Data Consortium, and D.S.S. contributed original research data or observations. J.K.B. led the writing of the manuscript, with substantial contributions from D.S.S., M.A.S., J.C.G., S.T.M., A.C.B., Z.S., C.D.G.H., J.L., M.T., S.E. and K.J.A.G. and editing by C.G.C. and the 2021 Western North American Heat Dome Data Consortium. All authors approved the final version of this paper.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41559-026-02987-6>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41559-026-02987-6>.

Correspondence and requests for materials should be addressed to Julia K. Baum.

Peer review information *Nature Ecology & Evolution* thanks Don Driscoll and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

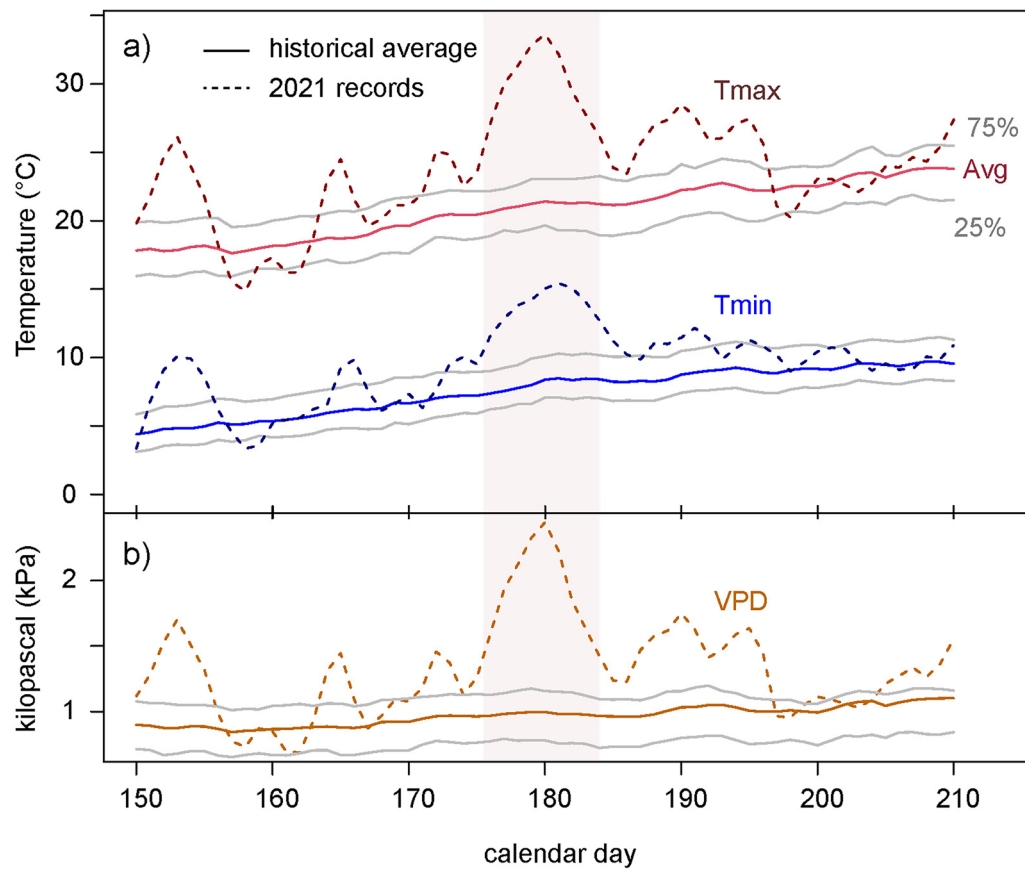
© The Author(s), under exclusive licence to Springer Nature Limited 2026

¹Department of Biology, University of Victoria, Victoria, British Columbia, Canada. ²Biodiversity Research Centre, University of British Columbia, Vancouver, British Columbia, Canada. ³Department of Zoology, University of British Columbia, Vancouver, British Columbia, Canada. ⁴Department of Botany, University of British Columbia, Vancouver, British Columbia, Canada. ⁵Department of Computer Science, University of Toronto, Toronto, Ontario, Canada. ⁶BC Ministry of Forests, Forest Ecosystems and Analytics Branch, Victoria, British Columbia, Canada. ⁷Department of Biology, Carleton University, Ottawa, Ontario, Canada. ⁸Department of Forest Resources Management, University of British Columbia, Vancouver, British Columbia, Canada. ⁹Institute for the Oceans and Fisheries, University of British Columbia, Vancouver, British Columbia, Canada. *A list of authors and their affiliations appears at the end of the paper. ✉ e-mail: baum@uvic.ca

The 2021 Western North American Heat Dome Data Consortium

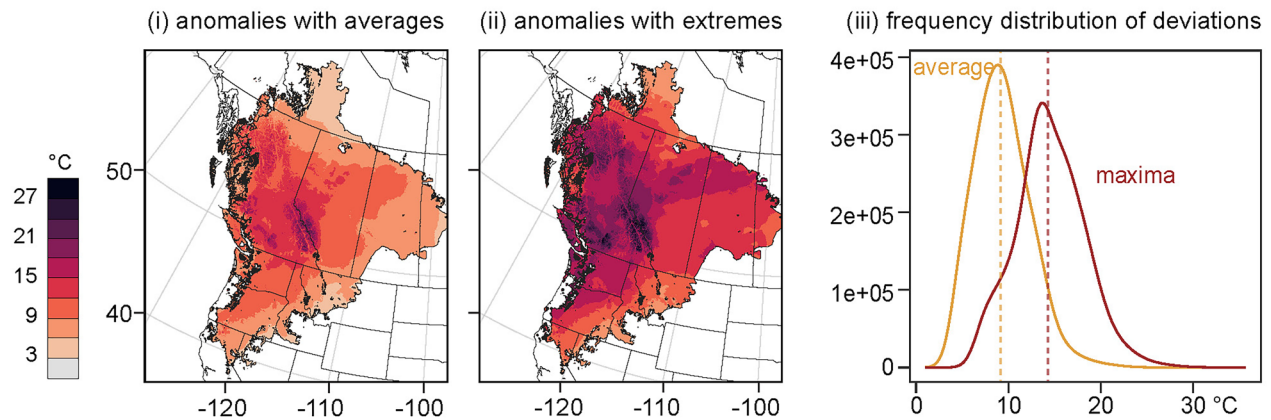
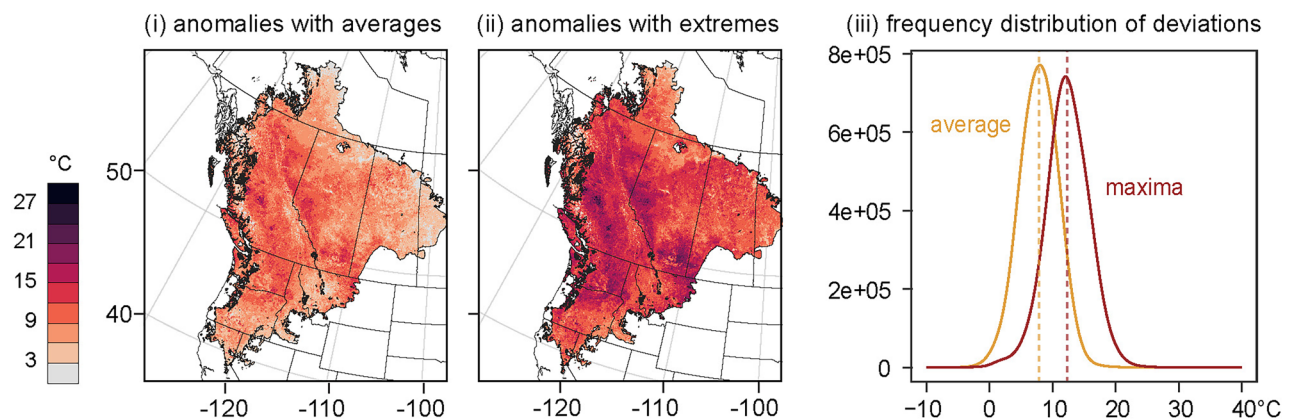
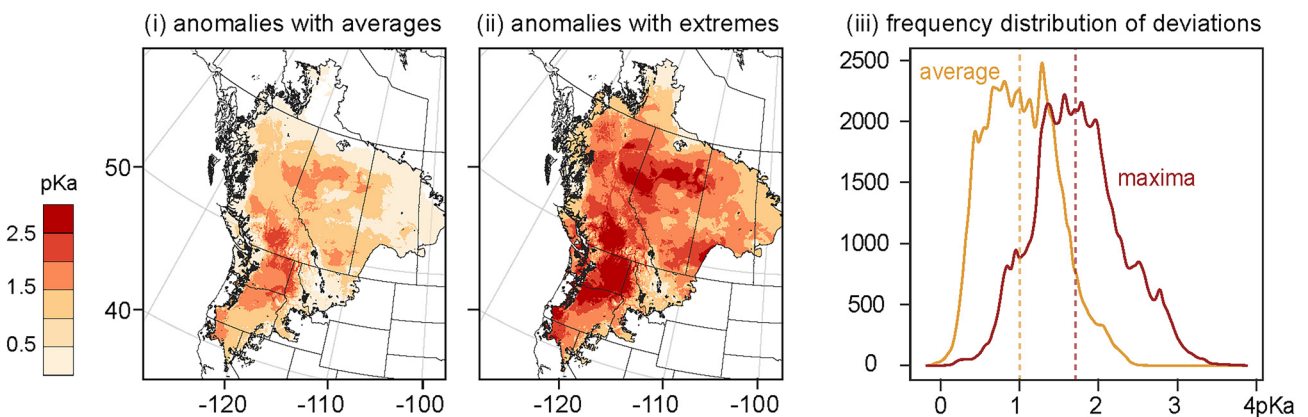
Paul K. Abram¹⁰, Amy L. Angert^{2,3,4}, Erin Bayne¹¹, Lori D. Daniels¹², Sarah Dickson-Hoyle¹², Heather Earle¹³, Michelle T. Franklin¹⁰, Alyssa-Lois M. Gehman^{9,13}, Tazarve Gharajehdaghipour⁷, Dexter Hodder¹⁴, Lionel Leston¹¹, Alison McAfee^{15,16}, Matthew G. E. Mitchell^{8,17}, Ken A. Otter¹⁴, Tamara Richardson¹⁸, Jordan Rosenfeld¹⁹, Gillian Sadlier-Brown¹³, Pablo Sandoval-Acuña^{2,3}, Carolyn Shores²⁰, Samuel Starko^{1,21}, Jason Thiessen¹⁰, Brian Timmer¹, Katie Tjaden-McClement⁷, Sunny Tseng¹⁴, Yonathan Uriel¹⁰ & Bert William²²

¹⁰Agriculture and Agri-Food Canada, Agassiz Research and Development Centre, Agassiz, British Columbia, Canada. ¹¹Department of Biological Sciences, University of Alberta, Edmonton, Alberta, Canada. ¹²Department of Forest and Conservation Sciences, University of British Columbia, Vancouver, British Columbia, Canada. ¹³Hakai Institute, Campbell River, British Columbia, Canada. ¹⁴Natural Resources and Environmental Studies, University of Northern British Columbia, Prince George, British Columbia, Canada. ¹⁵Department of Biochemistry and Molecular Biology, Michael Smith Laboratories, University of British Columbia, Vancouver, British Columbia, Canada. ¹⁶Department of Applied Ecology, North Carolina State University, Raleigh, NC, USA. ¹⁷Faculty of Land and Food Systems, University of British Columbia, Vancouver, British Columbia, Canada. ¹⁸Cornucopia Crop Consulting Ltd., Cawston, British Columbia, Canada. ¹⁹Aquatic Science Section, British Columbia Ministry of Water, Land and Resource Stewardship, Vancouver, British Columbia, Canada. ²⁰British Columbia Ministry of Water Land and Resource Stewardship, Williams Lake, British Columbia, Canada. ²¹School of Molecular and Life Sciences, Curtin University, Bentley, Western Australia, Australia. ²²Stuxwtéws (Bonaparte First Nation), Cache Creek, British Columbia, Canada.



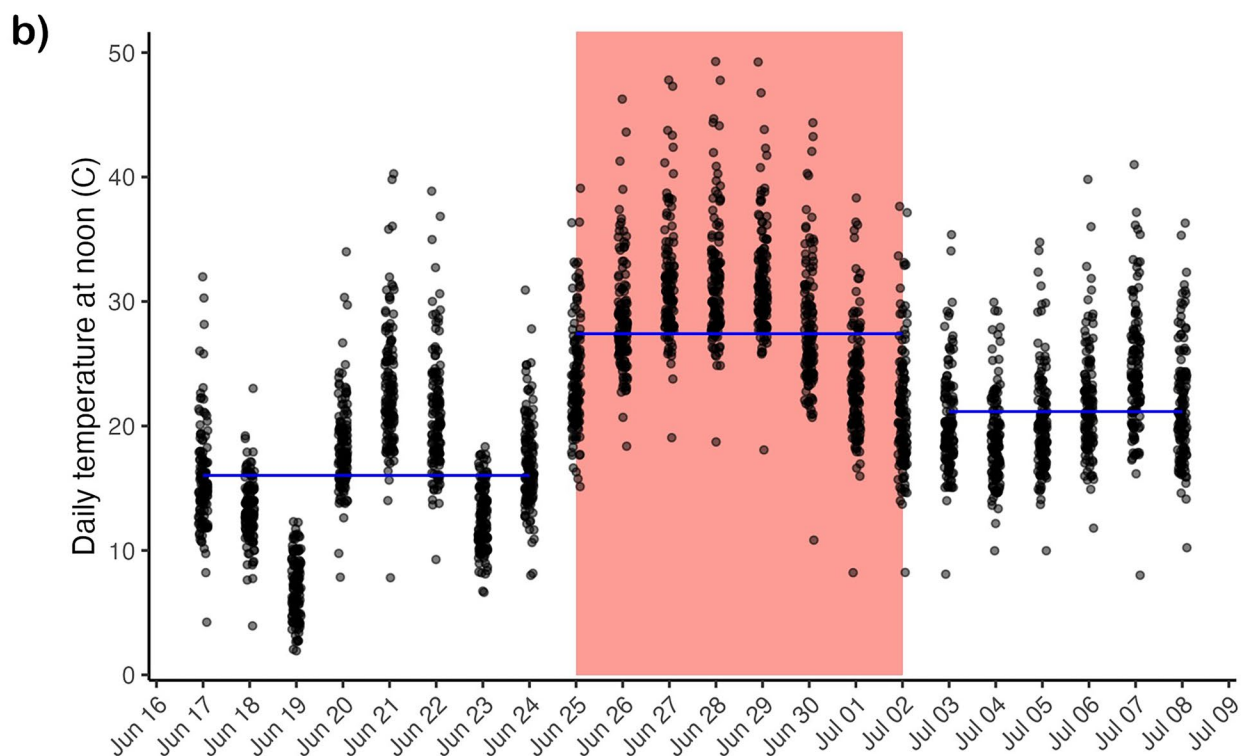
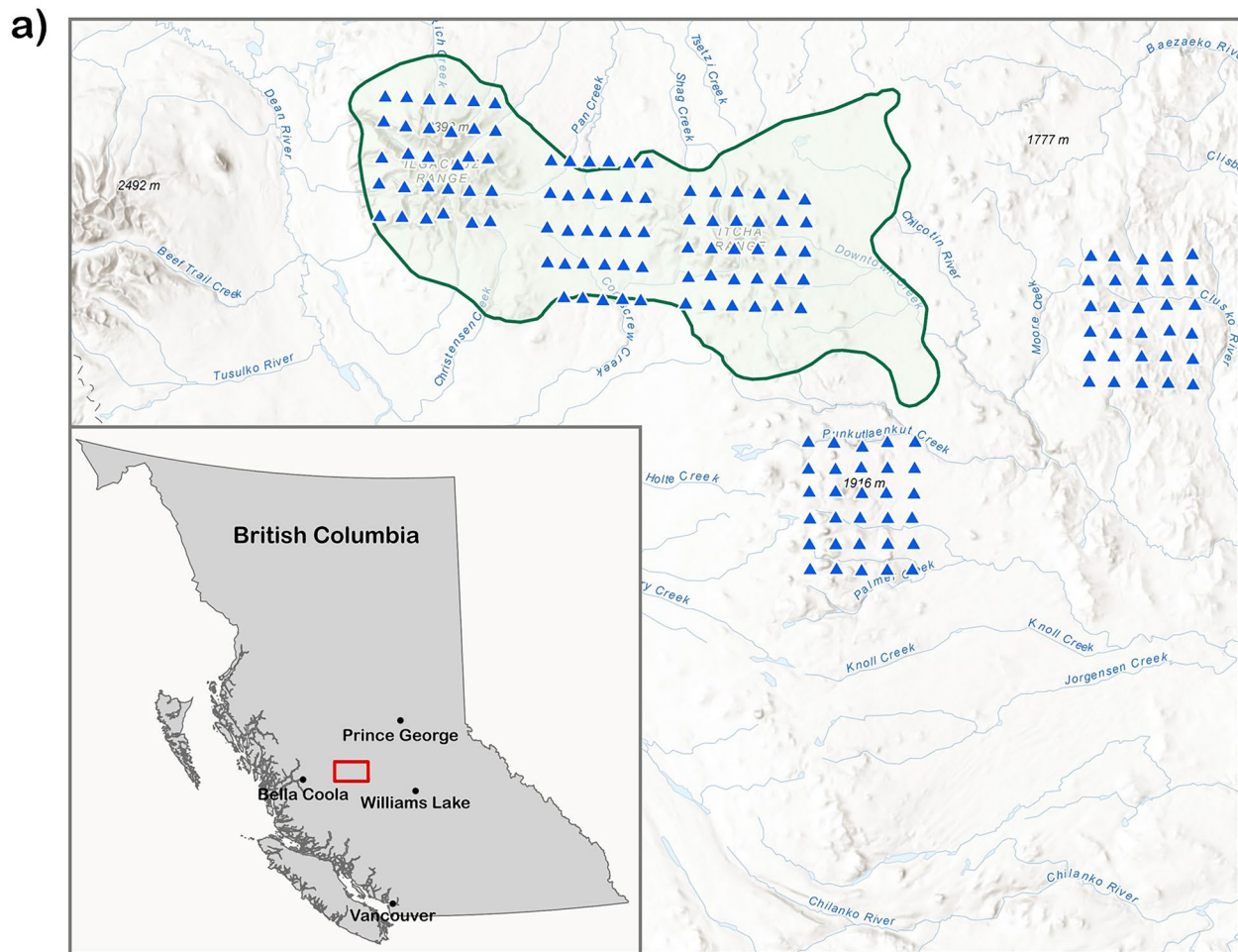
Extended Data Fig. 1 | Temperature and vapour pressure deficit (VPD) averaged across the region affected by the 2021 western North America heatwave for 2021 relative to the historical averages. a, Daily maximum (red) and minimum (blue) temperatures, and **b,** vapour pressure deficit (orange), averaged across the heatwave study region for 2021 (dashed lines) relative to the

historical (1980–2020) climatological mean (coloured solid lines) and 25th and 75th percentiles (grey solid lines). All three climatic variables show visibly higher values during the heatwave (June 25 (calendar day 177) to July 2 (calendar day 183)) shown in transparent light red shaded box.

a 2-meter air temperature**b Land surface temperature****c Vapor pressure deficit**

Extended Data Fig. 2 | Heterogeneity of thermal and evaporative stress anomalies during Western North America's 2021 heatwave. Deviations in **a**, 2-m daily maximum air temperature, **b**, daily day time land surface temperature, and **c**, heat-induced vapour pressure deficit are shown for the average of the heatwave days (left column) and extreme (that is hottest day of the heatwave;

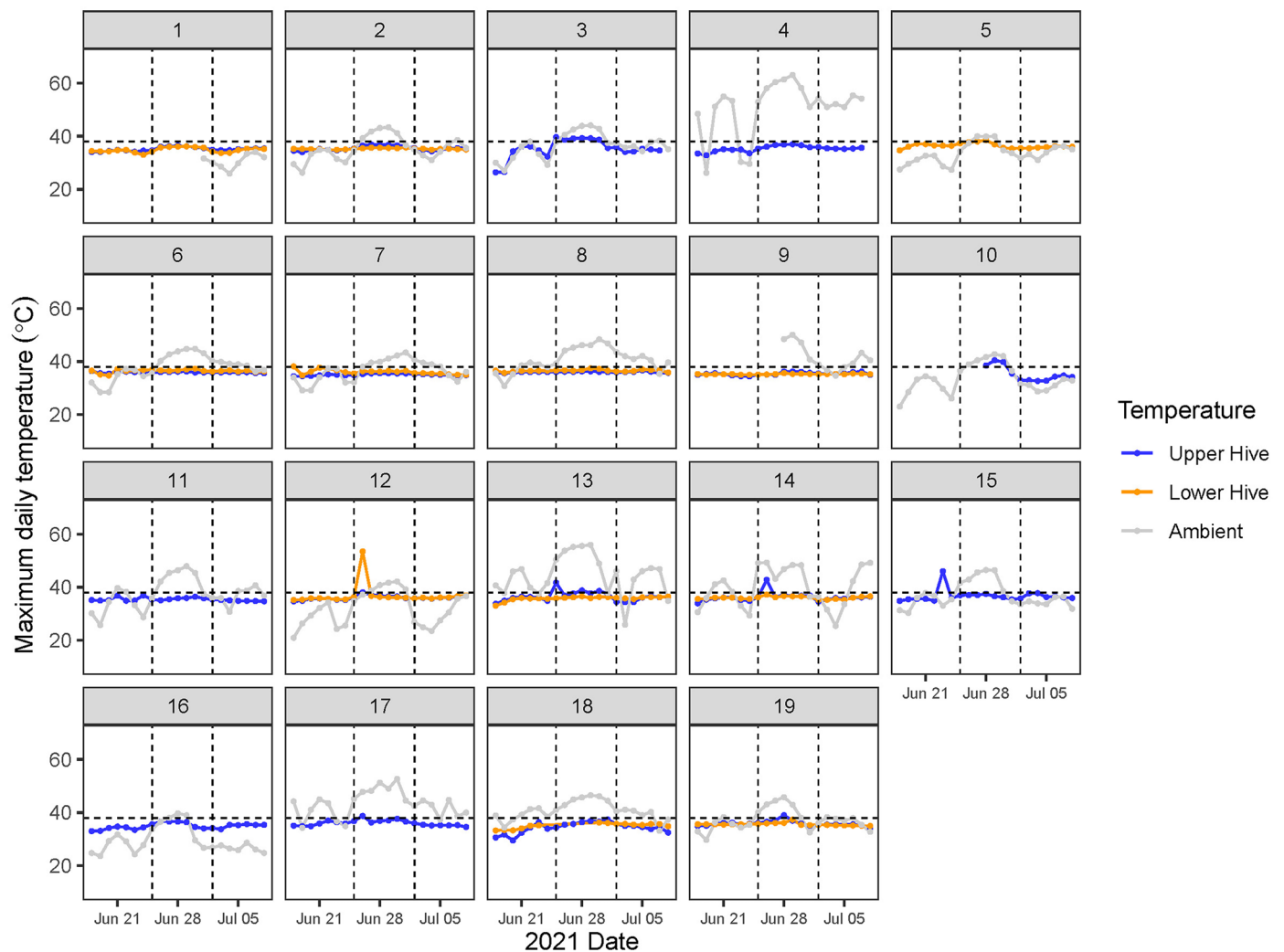
middle column) heatwave conditions compared to the historical (1980-2020) mean values in the corresponding heat dome period, June 25 to July 2. Deviations are calculated as 2021 values minus historical values. Frequency distributions for each variable shown (right column) along with median values (dashed vertical lines). Longitude and latitude are shown on all maps.



Extended Data Fig. 3 | Variation in daily temperature at noon before, during and after the 2021 heatwave in the Chilcotin Plateau, British Columbia.

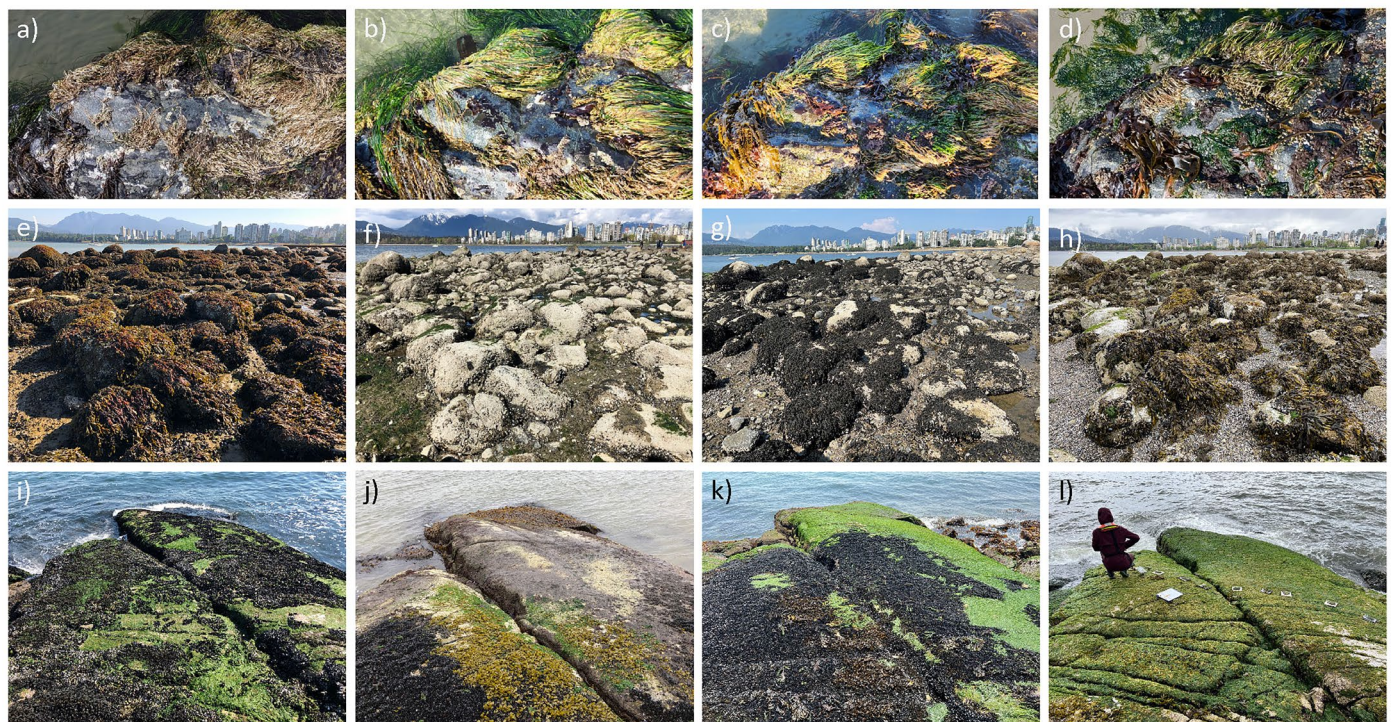
Temperatures were measured from 17 June - 8 July 2021 across $n = 149$ remote camera stations deployed to sample mammals in and near Itcha-Ilgachuz

Provincial Park in west-central BC - red square in inset (a). Black points represent temperature recorded at different camera stations, and blue lines denote mean temperatures for the periods before (left), during (red shading-center), and after (right) the heatwave (b).



Extended Data Fig. 4 | Temperature profiles of $n = 19$ honeybee hives in British Columbia. The British Columbia Honey Producers' Association runs a hive monitoring project, in which participants record and upload hive weight, humidity, internal temperature, and external temperature data at least hourly using the BroodMinder hive monitoring system. With the permission of the participants, we manually downloaded $n = 48,764$ data records for actively recording hives during the period of 18 June to –9 July 2021, from the archive at www.mybroodminder.com. Some beehives are made up of multiple boxes: blue

lines and dots denote temperature measurements from the upper hive box and orange lines and dots denote temperature measurements from the lower hive box. Grey dots and lines show the ambient temperature immediately outside the hives. The two black vertical dashed lines demark the start and end of the heatwave. For the majority of the hives the internal temperature of the hives remains below 38°C (horizontal grey dashed line), which is the temperature at which damage to queen's fertility is expected to occur¹⁰⁵ and damage to the developing brood is likely¹⁰⁶.



contemporaneous with
2021 heat dome

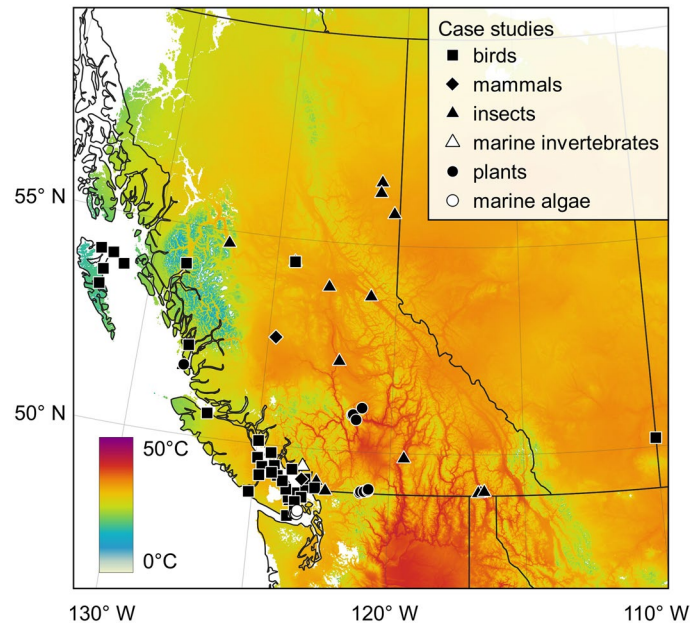
2022

2023

2024

Extended Data Fig. 5 | Photos illustrating intertidal impacts of the 2021 heatwave and subsequent recovery. Shown is recovery in a surfgrass (*Phyllospadix scouleri*) dominated permanent plot (**a-d**), a rockweed (*Fucus distichus*) dominated shore (**e-h**), and a mussel (*Mytilus trossulus*) dominated shore (**i-l**). **a**) Rocky outcrop in the lower intertidal zone covered in bleached surfgrass during the heatwave on June 29, 2021 near Odgen Point in Victoria, BC; **b**) Same site as in (**a**) one year later (June 30, 2022) with much healthier surfgrass. Surfgrass persists through 14 July 2023 (**c**) and 6 July 2024 (**d**), by which time kelp cover has increased in the plot as well. Note that the photo in (**d**) is slightly zoomed in. **e**) View looking northeast across the boulder shore at Kitsilano Point in Vancouver, BC shortly after the heatwave (July 4, 2021). Although the rockweed (*F. distichus*) and the mussels (*M. trossulus*) are still in place on this date, they have either already died or are in the process of dying - the reddish brown color in the *F. distichus* beds indicates tissue damage and/or necrosis; **f**) The same shore as in (**e**) on April 22, 2022, with very little recovery for *F. distichus* or *M. trossulus*. However, there are abundant barnacle (*Balanus glandula*) recruits coating the

rocks, representing an early stage of the successional process. **g**) By 7 July 2023, reproductively mature adult *B. glandula* are common. The percent cover of *F. distichus* is partially recovered with juvenile mussels present, but individuals of both species remain small. **h**) On 28 April 2024, recovery is ongoing, and mature size classes of both rockweeds and mussels are present. **i**) A granitic shore in West Vancouver during the heatwave (28 July 2021); although the mussel bed appears extensive, the majority of mussels in this photo have recently died yet their shells remain attached by byssal threads. **j**) The same site the following spring (18 April 2022), showing extensive bare space and a patch of surviving mussels (bottom left) that had received some protection during the heatwave by the orientation of the substrate and the presence of rockweed (visible in (**i**)). As of 6 July 2023 (**k**), small mussel recruits had reoccupied a substantial fraction of the area of the lost mussel bed, but by 4 June 2024 (**l**) these mussels had been consumed by sea stars during a period of uncharacteristically high salinity. Photo credits: **a-d**, B.T.; **e-l**, C.D.G.H.



Extended Data Fig. 6 | Location of ecological case studies included in the meta-analysis (n=32 taxa from studies). The coloured background areas, which were exposed to abnormally warm conditions during the 2021 heatwave, show

the average 2-m air temperature (colours indicated in scalebar) during the 2021 heatwave period, as in Fig. 1a ii. Longitude (horizontal axis) and latitude (vertical axis) are shown.

Extended Data Table 1 | Estimates from intercept only and final model after assessing collinearity amongst predictors and refitting estimates with cluster robust estimation methods

Coefficient	Estimate	Std. Error	T value	CI lower	CI upper
<i>Intercept only model</i>					
Intercept	-0.908	0.240	-3.791	-1.422	-0.394
<i>Final model</i>					
Intercept	-0.418	0.268	-1.563	-1.001	0.165
Time frame - during	-0.180	0.077	-2.349	-0.347	-0.013
Movement - sessile	-0.836	0.452	-1.851	-1.820	0.148

Bold text indicates coefficients with 95% confidence intervals (CI) that do not overlap with zero.

Extended Data Table 2 | Data sources for the n=32 taxa (n = 40 species total) included in the meta-analysis

Organism	Mobility/ Realm/ Response/ Control	Dates (B= before, D= during, A = after heatwave)	Measure; Method if not field counts	Location; Data providers (initials)
Algae and Plants				
Algae: Rockweed (<i>Fucus spp.</i>), Green rope seaweed (<i>Acrosiphonia spp.</i>), Sea lettuce (<i>Ulva spp.</i>), Red algae (Rhodophyta (Phylum)), Kelp (Laminariales (Order)); Surfgrass (<i>Phyllospadix scouleri</i>)	S/M/M/T; S/M/M,D/T	B (algae only): May 5-16, 2021; D: Jun 24-29, 2021; A: Jul 10-12, 2021	Cover (%); Photographs of defined plots	Victoria, BC; BT, SS
Arctic lupine (<i>Lupinus latifolius</i>)	S/T/R/T	B: Jul 27 - Sep 4, 2020; A: Jul 12 - Sep 12, 2021	Fruits per plant	Manning, Skagit Valley BC; KJAG, ALA
Deciduous shrubs: Prickly rose (<i>Rosa acicularis</i>), Saskatoon serviceberry (<i>Amelanchier alnifolia</i>)	S/T/D/S	A: Aug 20, 2021	Plants scorched w. no live tissue (%)	Southern Interior, BC; SD-H, LDD, BW
Surfgrass (<i>Phyllospadix scouleri</i>)	S/M/A,R/T	B: Aug 28-30, 2019; Aug 18-20, 2020; A: Aug 20-21, 24, 2021	Shoot & flowering shoot density (m ⁻²)	Calvert Is., BC; HE, GS-B, T. Froese, A-LMG
Birds				
Ferruginous hawk (<i>Buteo regalis</i>)	M/T/B/T	B: Jun 25-Jul 7, 2020; D: Jun 25 - Jul 2, 2021; A: Jul 3-7, 2021	Distance travelled /day (km); Radio telemetry	AB; EB, LL
Songbirds: Pacific wren (<i>Troglodytes pacificus</i>), Olive-sided flycatcher (<i>Contopus cooperi</i>), Western wood pewee (<i>C. sordidulus</i>), Ruby crowned kinglet (<i>Corthylio calendula</i>), Orange crowned warbler (<i>Leiothlypis celata</i>), Swainson's thrush (<i>Catharus ustulatus</i>)	M/T/B/T	B: Jun 15- 24, 2021; D: Jun 25-Jul 2; 2021 A: Jul 3-8; 2021	Detections/day; Open acoustic devices	Prince George, BC; KAO, ST, DH
Surf scoter (<i>Melanitta perspicillata</i>)	M/M/A/T	B: Dec 25, 1995 - Dec 25, 2020; A: Dec 25, 2021	Counts/party hour; citizen survey	Salish Sea, BC; Audubon (e-bird)
Insects				
Blueberry aphid (<i>Ericaphis fimbriata</i>), Raspberry aphid (<i>Amphorophora agathonica</i>)	S/T/A/T	B: Jul 5, 18, 2019 & Jul 3, 14, 28, 2020; A: Jul 7, 21, 2021	Plant terminals with aphids (%)	Abbotsford, BC; MTF, PKA, JT, YU
Western honeybee (<i>Apis mellifera</i>)	M/T/G/T	B: Jun 17- 24, 2021; D: Jun 25-Jul 2, 2021; A: Jul 2-9, 2021	Hive growth (Δ kg/kg/day); Hive mass monitor	Northern/ Central Interior, BC; AM
Mammals				
Bats (<i>Eptesicus fuscus</i> , <i>Myotis lucifugus</i> , <i>M. evotis</i> , <i>M. volans</i> , <i>M. yumanensis</i> , <i>M. californicus</i> , <i>Tadarida brasiliensis</i> , <i>Lasiurus cinerius</i> , <i>Lasionycteris noctivagans</i>)	M/T/B/T	B: Jun 25 - Jul 2, 2020; D: Jun 25 - Jul 2, 2021; A: Jun 25 - Jul 2, 2022	Total bat calls (all species); Open acoustic devices	Vancouver, BC; MGEM
Grey wolf (<i>Canis lupus</i>), Snowshoe hare (<i>Lepus americanus</i>), Caribou (<i>Rangifer tarandus</i>), Red squirrel (<i>Tamiasciurus hudsonicus</i>), Moose (<i>Alces alces</i>), Mule deer (<i>Odocoileus hemionus</i>)	M/T/B/T	B: Jun 17-24, 2021, D: Jun 25 - Jul 2, 2021; A: Jul 3-8, 2021	Daily detections; Camera trapping	Chilcotin Plateau, BC; ACB, KT-M, TG, CS
Marine Invertebrates				
Bay mussel (<i>Mytilus trossulus</i>), Acorn barnacle (<i>Balanus glandula</i>)	S/M/M/S	A: Jul 8, 2021 (<i>M. trossulus</i>), Aug 19, 2021 (<i>B. glandula</i>)	Percent mortality	Salish Sea and Central coast, BC; CDGH
Dogwhelk (<i>Nucella lamellosa</i>)	M/M/M/T	B: Apr 19, May 20, 2011; Apr 9, May 9, 2012; Apr 27, 29, May 12, 2013; May 14-16, 2014; A: Jul 11, 22, 2021	Percent mortality	Salish Sea, BC; CDGH
Thatched barnacle (<i>Semibalanus cariosus</i>)	S/M/M/S	A: Jul 2021	Percent mortality	Salish Sea, BC; A. Hesketh, CDGH
Barnacles: Thatched (<i>S. cariosus</i>), Acorn (<i>Balanus glandula</i>), Little brown (<i>Chthamalus dalli</i>)	S/M/A/S	A: Jul 8, 2021 (<i>B.g.</i> & <i>C.d.</i>); Aug 21, 2021 (<i>S. c.</i>)	Number of recruits	Salish Sea, BC; A. Hesketh, CDGH

These are ordered by broad taxonomic group, including study location (BC=British Columbia, Canada, AB=Alberta, Canada, depicted in Extended Data Fig. 3), data collection dates, response types (M = mortality, G = growth, R = reproduction, A = abundance, D = damage, B = behaviour), measurement type and method, data providers, control type (T = temporal, S = spatial), and two categorizations examined in the meta-analysis: motility (motile (M) vs. sessile (S)) and realm (marine (M) vs. terrestrial (T)). Given their close connection, mortality and abundance were combined into a single response type, P=persistence, in the meta-analysis.

Extended Data Table 3 | Evidence for ecological effects of 2021 heatwave not included in the meta-analysis across a breadth of taxa (n = 25 species total, data for 2 of the 25 species from other sources were included in the meta-analysis), ordered by broad taxonomic grouping (algae, birds, fish, insects, mammals, marine invertebrates, plants)

Taxa	Ecosystem; Location	Method	Impact	Type	Details, with References and Source Type
Algae					
Rockweed (<i>Fucus distichus</i>)	Intertidal; Southern BC coast	Opportunistic observations post HW	↓	Mortality	Extensive rockweed tissue damage and subsequent mortality; CDGH and SE direct observations in 2021. (O)
Birds					
Bushtit (<i>Psaltirparus minimus</i>)	Urban park; Portland	Long-term nest monitoring study	↓	Mortality	During the 2021 heatwave 8 active bushtit nests were monitored. Nestling mortality was 100% in 3 nests and 4 broods survived at least in part ⁴⁰ . (L)
Raptors	Various; BC south to OR	Online media reports of rehabilitation centres	↓	Injury & mortality	Rehabilitation centres received hundreds of raptors, including Swainson's hawks Cooper's hawks, osprey and merlin, during the heatwave. Birds jumped from nests 60ft high due to overheating causing many injuries including broken legs and multiple wing fractures ^{108,109} . (M)
Multiple species	Lynwood, WA	Online media report of rehabilitation centres	↓	Injury & mortality	Caspian tern (<i>Hydroprogne caspia</i>) chicks died or sustained injuries after leaping from hot roof in Seattle; baby swallows, sparrows, raptors (Cooper's hawks, bald eagles) were brought to the rehabilitation centre after jumping from their nests ^{110,111} . (M)
Fish					
Salmonids	Stream; Aldergrove, BC	Observed during scheduled field work (during HW)	↓	Morbidity and mortality	Unusually high stream temperature (23-24C); Several dozen dead juvenile coho and trout observed over a representative 200m reach; fish seen gasping for air at the stream surface; JR direct observation in 2021. (O)
Insects					
Aquatic insects	Urban tree-holes; Southern BC	Recorded in 2021 (N=47), 2022 (N=101)	No effect	Larval mortality	Very little (<1%) mortality was observed one week after the 2021 heatwave, and similar in magnitude to 2022 (ANOVA, F=1.93, p =0.17) ¹¹² . (L)
Marine Invertebrates					
Shellfish	Coastal; Vancouver, BC	Post 2021 heatwave report	↓	Disease	Following the 2021 heatwave high sea surface temperatures facilitated the perfect conditions for vibrio bacteria to contaminate shellfish. This led to over 50 people contracting vibriosis illness ¹¹³ . (M)
Little brown barnacle (<i>Chthamalus dalli</i>), kelp crab (<i>Pugettia productus</i>), jingle shell (<i>Pododesmus macroschisma</i>), leather star (<i>Dermasterias imbricata</i>), ochre sea star (<i>Pisaster ochraceus</i>)	Intertidal; Salish Sea, BC	Opportunistic observations (during and after heatwave)	↓	Mortality	Extensive barnacle mortality, occasional freshly dead ochre stars, burnt looking leather stars on sun exposed rocky shores. Recently dead kelp crabs and jingle shells. Most sea stars were able to find shade; little mortality was observed; CDGH direct observations in 2021. (O)
Giant California sea cucumber (<i>Apostichopus californicus</i>)	Subtidal; Nanoose Bay, BC	Community science	↓	Mortality (disease)	Extensive reports of Giant California sea cucumber wasting leading to mortality (up to 94% of observations). Higher mortality in shallow subtidal ¹¹⁴ . (L)
Plants					
Seagrass	Intertidal seagrass meadow; San Juan Islands, WA	Multi-year UAV survey	↓	Mortality	In all monitoring sites there was an immense loss in density and coverage of the foundation species, seagrass (<i>Zostera</i>). Beach haven site saw a near complete loss of upper 10m of seagrass between 2020 and 2021 ¹¹⁵ . (L)
Salal (<i>Gaultheria shallon</i>)	Forest; Maple Ridge, BC	Opportunistic observations post HW	↓	Damage	Extensive leaf scorch on salal, a shrub in the Malcolm Knapp Research Forest following the 2021 heatwave; DSS direct observation in 2021. (O)
Douglas fir, Western hemlock trees	Plantation forest; Western OR & WA	Aerial detection surveys	↓	Damage	>90,000 ha of foliar damage following heatwave with the most extensive impacts in coastal Douglas fir & western hemlock plantation forests on ¹¹⁶ . (M)
Cherry trees; spotted wing drosophila	Organic agriculture Kelowna, BC	Observed during HW-scheduled field work	↓	Mortality	Cherry crops severely damaged by heatwave ¹¹⁷ . No Spotted Wing Drosophila—a key pest of these crops in other years—emerged from damaged cherries; T.R. direct observation in 2021. (M, O)

Sources include peer-reviewed literature^{107–117} (L) (C.D.G.H. and S.E., direct observations in 2021; J.R., direct observations in 2021; C.D.G.H., direct observations in 2021; D.S.S., direct observations in 2021; T.R., direct observations in 2021), field observations (O) by authors of this paper, and online media (M) reports. Raw data and analysis for quantitative field observations are presented within the repository for this article (see Data Availability and Code Availability statements); qualitative observations are reported in their entirety here.

Extended Data Table 4 | Candidate models predicting the effect of temperature and meltwater potential on stream flow during and after the 2021 heatwave

			June			August		
Model	df	log L	AICc	delta		log L	AICc	delta
intercept only	2	-2.4	8.9	169.1		135.2	-266.4	38.9
glacier	3	27.5	-48.9	111.3		149	-291.8	13.5
snow	3	63	-119.9	40.3		144.5	-283	22.4
surface	3	2.8	0.5	160.7		143.4	-280.6	24.7
surface + glacier	5	33.1	-55.9	104.3		157.8	-305.2	0.1
surface + ice	5	19.7	-29.1	131.1		149.6	-288.8	16.5
surface + ice + glacier	9	46.1	-73.2	86.9		161	-302.9	2.4
surface + snow	5	68.3	-126.3	33.9		152	-293.7	11.6
surface + snow + glacier	9	72.6	-126.1	34		162.1	-305.2	0.2
surface x glacier	4	30.5	-52.9	107.3		155	-301.8	3.5
surface x ice	4	18.4	-28.5	131.7		149.6	-290.9	14.4
surface x ice x glacier	5	39.2	-68.1	92.1		157.8	-305.3	0
surface x (ice + glacier)	7	42.7	-70.8	89.4		159.8	-305	0.3
surface x snow	4	66.5	-124.8	35.4		151.1	-294	11.4
surface x snow x glacier	5	70	-129.6	30.5		155	-299.7	5.6
surface x (snow +glacier)	7	71.1	-127.6	32.5		157.9	-301.2	4.1
air	3	26	-45.8	114.4		135.8	-265.5	39.8
air + glacier	5	57	-103.8	56.4		149.6	-288.9	16.5
air + ice	5	48.8	-87.3	72.9		146	-281.6	23.7
air + ice + glacier	9	73.5	-128	32.2		157.2	-295.3	10
air + snow	5	85.3	-160.2	0		144.9	-279.4	25.9
air + snow + glacier	9	89.1	-159.1	1.1		152.7	-286.3	19
air x glacier	4	46.7	-85.2	75		149.2	-290.1	15.2
air x ice	4	47	-85.8	74.4		143	-277.7	27.6
air x ice x glacier	5	60.3	-110.2	49.9		152.2	-294.2	11.2
air x (ice + glacier)	7	72.1	-129.6	30.6		154.9	-295.1	10.3
air x snow	4	77.4	-146.6	13.6		144.7	-281.2	24.1
air x snow x glacier	5	80.7	-151	9.2		149.2	-288	17.3
air x (snow + glacier)	7	86.5	-158.4	1.8		149.7	-284.7	20.7

Response variables were the ratio of streamflow in 2021 to that averaged over 2000-2020 in each stream (n=212), either during the 2021 heatwave period ("June" models: June 25 - July 2) or later that summer ("August" models: August 1-31). Explanatory variables included: glacier = proportion of catchment covered by glaciers, square-root transformed; snow = proportion of pixels with >40% snow cover June 22-26 2021, square-root transformed; ice = snow cover calculated just for glaciated part of catchment; surface = temperature anomaly of mean daily maximum surface temperatures in the period June 25-July 2 (2021 versus the previous 40 years); air = deviation of the mean daily maximum air temperature in the 2021 heatwave period (June 25 - July 2, 2021) from the week prior (June 17-24, 2021). All models were generalized linear models fitted with Gaussian errors and assessed with log likelihood ("log L"). Model selection for both June and August models was based on AICc, with selected models (indicated with boldface) having either the lowest AICc or within 2 AICc units of the lowest AICc (delta ≤ 2). Data was compiled from online sources as described in Materials and Methods.